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Electric vehicles development in Sub-Saharan Africa: Performance assessment of standalone renewable energy systems for hydrogen refuelling and electricity charging stations (HRECS)

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ABSTRACT

One of the main setbacks to electric vehicles (EVs) adoption relates to the unavailability of the required charging stations across a jurisdiction. In addition, drawing the required energy of the EVs from a fossil fuel-dominated grid does not only minimize the cleaner benefits of EVs but also puts enormous pressure on an already stressed grid network in electricity-deficient regions like Sub-Saharan Africa (SSA). Hence, in the current study, six different 100% hybrid renewable energy systems based on powerplants of solar, wind, and biomass have been proposed for meeting the energy needs of 70 and 30 battery and fuel cell electric vehicles, respectively using Ghana as a case study. With the aid of HOMER Pro software and multicriteria decision-making tools, the hybrid system, biogas generator-photovoltaic emerged as the most feasible of the six proposed solutions from a technical, economic, and environmental standpoint. This winning system produces 3.9 GWh/yr and 55.6 tonnes/yr of electricity and hydrogen, respectively. The NPC, LCOE, LCOH, and payback period was recorded as \$6.53 million, \$0.52/kWh, \$9.09/kg and ~8 years, respectively. Approximately, 4 tonnes of GHG emissions are produced by the HRECS each year, and the amount of CO_2 emissions avoided from replacing 100 gasoline vehicles with the proposed EVs has been obtained as 460 tons/yr. With the aid of sensitivity analysis, it became apparent that the feasibility of the proposed systems could improve with improvement in components' efficiencies and lifetime, and reduction in unit costs. The current work has several contributions to the sustainable development goals (7, 8, 11, 13), and could aid in accelerating EV penetration in SSA.

1. Introduction

Throughout the world, an efficient transport sector is at the backbone of socio-economic development and human activities. By extension of this provision of service, the transport sector has, over the years, evolved into the world's second-largest greenhouse gas (GHG) emitter, with road transportation currently responsible for at least 90% of emissions from the sector (Tongwane &Moeletsi, 2021). In Sub-Saharan Africa (SSA), the transport sector plays an important role in the energy system, and the demand for fossil fuels by the sector continues to surge as the population and economic activities within the region increase (Collett et al., 2021). In 2018, the SSA transport sector demanded about 15% of the final energy consumption (International Energy Agency (IEA) 2019), and the increase in demand for high carbon fuels (diesel and gasoline) poses environmental and health risks to its citizens. Vehicle emissions are partly responsible for ambient air pollution and the release of high amounts of particulate matter (PM2.5), which gradually deteriorates air quality (Booysen et al., 2022). In 2015, the PM and ozone from the transport sector were estimated to have contributed to nearly 4500 deaths in SSA (Anenberg et al., 2019).

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Electric vehicles (EVs) have gained a global momentum due to their cleaner aspects compared to conventional vehicles operating on the internal combustion engine (ICE). The adoption of EVs provides a huge avenue for the reduction of GHG emissions from the transport sector (Woo et al., 2017). A recent study in South Africa estimates that electric motorcars, for example, have the potential to reduce baseline GHG emissions of motorcars by 19% in 2050, whereas conventional motorcars will contribute 63% (Tongwane &Moeletsi, 2021). Given the global transition toward low or zero-carbon transport systems, the adoption of EVs in SSA is inevitable. Also, when EVs are adopted on a large scale, many governments in SSA could divert the high fossil fuel subsidies into financing affordable and green electricity (Collett et al., 2021). Furthermore, EVs are highly efficient compared to conventional vehicles; an EV is capable of converting up to about 60% of the grid electricity into motive power at the wheels, whiles a typical conventional gasoline vehicle can convert up to about 20% of the stored energy in gasoline to power at the wheels (Energy Commission n.d.). EVs are cheaper to maintain and refuel and produce much lower noise pollution compared to ICE vehicles.

Ghana is one of the seven countries responsible for about 70% of Africa's transport emissions (Ayetor et al., 2021). The Ghanaian transport sector in 2016 emitted 7.3 million tons of carbon dioxide (CO_2) emissions, of which 92% were borne by the road transport sector, representing 52% of the country's total CO_2 emissions (Ackaah et al., 2021). In a recent state of the environment report, the transport sector has been identified as the largest source of air pollution in Ghana (Environmental Protection Agency, 2018). By virtue of their propensity to cause adverse human health, some key pollutants from the country's transport sector are air borne particulates (PM_{10} and $PM_{2.5}$). These airborne particulates are highly linked to acute respiratory illness (ARI) and chronic lung disease (CLD) which are ranked in the top 5 major causes of deaths in Ghana (Angnunavuri et al., 2019). As a signatory to the 2015 Paris Agreement and Sustainable Development Goals (SDGs), Ghana has updated its commitment to generating absolute GHG emission reductions of 64 $MtCO_2e$ by 2030 (MESTI, 2021). For the realization of this target, there is the need for the development of a wide range of policy actions, including but not limited to a sustainable road transportation sector powered by low or zero-carbon fuels. A recent study by World Health Organization (WHO) has found that if Ghana were to adopt sustainable modes of transport in its capital city, Accra, the improvement in air quality could prevent up to 5500 premature deaths and averted healthcare costs of about USD 15 billion (WHO 2021). Against this backdrop and in conjunction with the associated health benefits, Ghana recently became a member of the Global Fuel Economy Initiative (GFEI), an initiative with a primary aim of reducing fossil-fuelled vehicles whiles gradually introducing EVs and other sustainable transport mode systems (Angnunavuri et al., 2019). The GFEI has benefits including but not limited to reduction in urban air pollution, over USD 300 billion and USD 600 billion of fuel savings by 2025 and 2050, respectively, and 1–2 Gt/yr of CO_2 emission reduction by 2025–2050 (CCAC NPA UNEP, 2016). Due to the country's relatively higher electricity access rate (85%), Ghana is one of Africa's seven countries with a greater chance of EV penetration (Ayetor et al., 2022).

Despite the huge benefits, growths, and prospects, the transition toward EV is met with several challenges (Fig. 1) (World Economic Forum, 2018). One of the most important barriers to EV rollout is the provision of the electricity required for charging; depending on the existing grid to meet the increasing demand of EV charging stations has several inherent problems, including extra peak loads by the system. The situation ultimately results in violation of voltage limits, transformer and feeder overloads at different voltage levels, and negative impacts on the grid (Al Wahedi and Bicer, 2022). Alternatively, Ghana has abundant and readily available renewable energy sources (RES) such as solar, wind, biomass, and mini/medium hydro that could be harnessed for such additional electrical loads (Afrane et al., 2022). A 35 EJ of solar potential has been estimated for Ghana (Eshun &Amoako-Tuffour, 2016); 39 exploitable sites have been identified as feasible for mini and medium hydro plants with a potential of 800 MW (Energy Commission Ghana, 2012; Essel, 2015); the wind energy potential along the eastern border with Togo is estimated at around 300 MW capacity or 800 GWh of electricity (Ghana Premium Consultant, 2016); 35.88 MWp and 27.60 MWp of electricity could be generated from the municipal solid waste (MSW) produced in Accra and Kumasi, respectively (Fiagbe, 2020) whiles crop and animal dung in the country has been estimated at bioenergy potentials of 47–100 TJ (Duku et al., 2011). Due to the huge renewable energy (RE) potential in the country, as indicated supra, supplying the charging demand by RES will not only meet the required demand reliably but relieves potential pressure from the grid and is also an eco-friendly strategy for electricity generation.

There are several electric vehicles available in the market, including battery electric vehicles (BEVs), hybrid electric vehicles (HEVs), plug-in hybrid electric vehicles (PHEVs), and fuel cell electric vehicles (FCEVs). Existing studies have conducted several feasibility studies for renewable energy-based charging/refuelling stations, mainly charging stations for battery EVs in Bangladesh (Podder et al., 2022), Qatar (Al Wahedi and Bicer, 2022), Brazil (Schetinger et al., 2020), Vietnam (Minh et al., 2021), or refuelling stations for fuel cell EVs in China (Chen et al., 2021), Tukey (Gökçek &Kale, 2018), South Africa (Ayodele et al., 2021), Sweden (Siyal et al., 2015), Brazil (Micena et al., 2020). However, studies related to a hybrid station for both BEV charging and FCEV

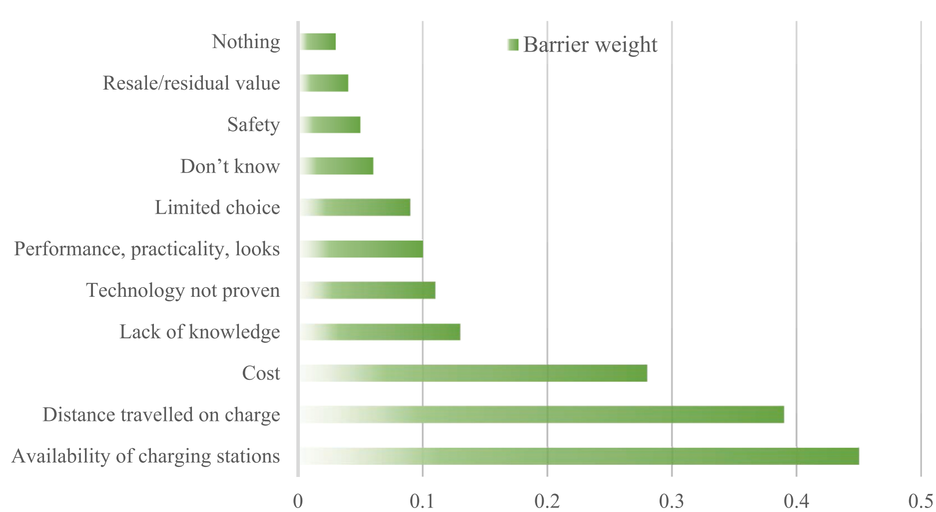

Fig. 1. Prioritization of barriers to EV adoption by a pool of potential customers (World Economic Forum, 2018).

refuelling are relatively scarce in the extant literature. Furthermore, these studies either rely on the objective function of the modelling software usually net present cost or one or two user-defined objective functions to select the most optimal hybrid system. But in reality, there are several factors ranging from technical, economic, environment, and to some extent social and political that are very critical in deciding the feasibility of hybrid systems, and this is a multi-criteria decision-making problem which has not been considered by the mentioned studies. In addition, most of these studies have been conducted for areas within Europe, America, and Asia, with that of SSA almost non-existent. The renewable energy sources also considered for these studies are mostly one or with only few studies considering two types. But it is known that hybrid energy systems based on multiple RES in a particular locality is vital for addressing the volatility and randomness of a single RES (Wang et al., 2022).

Hence, in the current work, (1) a novel configuration is set up consisting of different RES, including solar, wind, and biomass, to simultaneously provide electricity and hydrogen to the two types of EVs at the same location. The design is named a 'hydrogen refuelling and electricity charging station' (HRECS) for FCEVs and BEVs, respectively, powered by the selected RES. HRECS is expected to meet the daily demand of 70 BEVs and 30 FCEVs with battery/tank capacities of 35 kWh and 5 kg per EV, respectively, in Accra, Ghana. To the best of the authors' knowledge, this will be the first work with such a configuration specifically modelled for a developing region, in this case, a Sub-Saharan African country. Other contributions made in the current study include; (2) The selection of optimal energy systems among several alternatives can often be complicated and presents a multicriteria decision problem. In such instances, there is the need to utilize appropriate tools with the capability of identifying and ranking these alternatives in order of viability, considering technical, economic, and environmental factors. Therefore, the multicriteria decision analysis (MCDA) models CRITIC and PROMETHEE II have been selected and used in the current study to reveal the best RE system for HRECS. (3) The profitability index, which is an economic indicator for RE projects, has been included to ascertain whether the proposed project is worthy of pursuit. (4) Furthermore, a breakeven grid extension distance is calculated to obtain the optimal distance above which standalone RES for HRECS is preferable over grid extension and vice versa. (5) Grid-connectivity has been analyzed to compare its feasibility against the proposed standalone system for HRECS in the selected location. (6) A sensitivity analysis has also been conducted in the present study to test the robustness of the original findings when subjected to changes in components' cost, fuel cost, grid sale capacity, electricity sell back rate, discount rate, expected inflation rate, and components' efficiency and lifetime.

In line with the Paris Agreement and the UN's SDGs, immediate policy actions are required across SSA for the decarbonization of the region's transport sector, especially road transportation. It has been established in this section that transition towards EV is one of the most important and feasible pathways for the realization of these targets. By carefully designing an off-grid charging/refuelling station powered by RES, both BEVs and FCEVs available in the region can be charged/refuelled reliably in an eco-friendly manner. The novel hybrid energy systems designed in the current work for the region's road transport sector has several contributions to SDGs including but not limited to; (1) zero or low carbon technologies with environmentally feasible characteristics that will be pivotal in delivering on Ghana's EV policies and strategies including required charging/refuelling infrastructure (contribution to SDG7a: enhance international cooperation to facilitate access to clean energy research and technology); (2) the use of low carbon technologies such as EVs results in the reduction in GHG emissions in transportation sector, enabling the integration of climate change measures in national strategies and planning (Contribution to SDG13: Climate Action 13.2 - Integrate climate change measures into national policies, strategies and planning); (3) EVs and other sustainable transport modal systems reduce local air and noise pollution in cities, ensuring safe and resilient cities (Contribution to SDG11: Making cities resilient and sustainable); (4) One social benefit in the development of RE sector is the creation of jobs. The use of RES as a power source for charging/refuelling infrastructure unlocks job opportunities, boosting the employment rate of the region (Contribution to SDG8: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all).

The remaining parts of the current study is organised as follows: Section 2 presents the current state of EV adoption and its potential challenges in West Africa with more emphasis on Ghana; in Section 3, prospective readers are briefed on some of the latest studies related to the current study from different locations with a summary of their objectives, research design and key findings; Section 4 is the method section and it provides an overview of the selected case study area (climate and resource assessment), adopted tool (HOMER Pro), and key mathematical models of assessment; Section 5 describes the steps taken to rank the six proposed hybrid systems by MCDA models; the obtained results are discussed comprehensively in Section 6; results from the sensitivity analysis is contained in Section 7; finally, the current research is concluded in Section 8.

2. Status of EV adoption in West Africa and potential challenges

Compared to any other continent, Africa has the highest urbanization rate at a 4% annual growth as opposed to the global 2%. Consequently, this rapidly increasing urban population puts pressure on existing infrastructure and resources – particularly transport and energy (Hill et al., 2018). Despite the significant growth of EVs on the world's roads (16.5 million by the end of 2021, i.e., about thrice 2018's) (IEA, 2022), only a small share of total vehicle sales in Africa are EVs. The adoption and prioritization of EVs by West African countries, in particular, has been abysmal compared to other SSA countries such as South Africa, Kenya, and Rwanda. However, few West African countries in the region, the likes as Cape Verde, Togo, Nigeria, and Ghana, have indicated their interest in EV adoption for their road transport. A 100% EV target for new sales of passenger cars by 2035 and urban buses by 2040 has been set out by Cape Verde (Khan et al., 2022). In Togo, plans are underway to produce 1000 EVs (motorbikes and three-wheelers) per day for the local market. The state has also made provisions for the exemptions of EVs from import duties (Wansi, 2022). Last year, the first EV made in Nigeria (the Hyundai Kona) was unveiled (Dioha &Caldeira, 2022).

The number of vehicle owners in Ghana has seen a tremendous rise since the start of 2000. The available statistics show a rise in vehicle fleet by over 300% from 2000 (511,7755 vehicles) to 2016 (2,098,726 vehicles) (EPA, 2017). The largest share of registered vehicles is concentrated in the capital region of the country. Also, the ratio of vehicle to population has witnessed a steady growth since 2010, from about 50 vehicles per 1000 population to about 70 vehicles per 1000 population in 2015 (CCAC NPA UNEP, 2016). The EV market in Ghana is gradually evolving, and interesting trends are expected in the near foreseeable future. There are some EV developments already ongoing in the country, which are briefly presented as follows. In October 2019, the Energy Commission, in partnership with the Ministry of Energy, rolled out a 'Drive Electric Initiative' with the sole aim of promoting EVs on Ghana's roads. Over 100 EVs and a minimum of 10 public charging outlets were set to be in place in Ghana by 2020. Also, the Kona EV model of Hyundai was launched in Ghana in 2020. The Ora Black Cat introduced to the Ghanaian market by Egle Motors Ltd in early 2021 has already sold over 20 units. Similarly, more than 15 different EVs have been offered on lease or outright purchase by SolarTaxi. Accraine, which is a Ghanaian firm with offices in both Ghana and the United Kingdom, has also begun supplying BEVs and PHEVs to customers in Ghana. Local manufacturer, Katanka also introduced the 'Amoanimah' EV in Ghana in 2020. In a recent announcement this year by the Ministry of Transport, the government intends to pilot electric buses in the country as part of the continued efforts to decarbonize and achieve net-zero emissions (NZE). The Ministry is therefore on the verge of importing ten electric buses to selected cities for the pilot program. From the infrastructure point of view, POBAD International, in collaboration with the Electricity Company of Ghana, launched EV public charging stations in the country (Forgor, 2022; GFEI, 2016; Kuhudzai, 2021).

The gradual EV revolution in West Africa will, however, introduce new challenges. In an assumed 2050 scenario where 70–75% of all road transport modes in the region are EVs, about 9.9 million EVs could be on the roads in West Africa. Under this scenario in 2050, the electricity demand of these EVs in the region could reach about 105,000 GWh, representing about twice the final electricity demand of the region in 2015. An additional USD 20 billion (excluding costs of charging infrastructure) would be required to meet this demand through investment in generation and grid infrastructure. Furthermore, since EVs are likely to be concentrated in urban parts of the countries within the region, the rise in electricity demand for the EVs will further stress the already constrained electricity grid (Nana, 2021). The situation presents an opportunity for harnessing the RES in the region for standalone or grid-tied systems to supply the energy demands of the EVs in the region. There is also the benefit of providing power to the grid where excess RE electricity from the charging station is injected into the grid network for other energy demands such as residential, industrial, commercial, and agriculture. The strategy will allow the increase in electricity access rate in the region, where more than half of the population (58% (Puliti, 2022)) have no access to electricity.

Against this backdrop, it is important for the feasibility study of such standalone or grid-tied RE infrastructure for EV charging in West Africa.

3. Literature review and state-of-the-art of EV field

In recent years, studies on BEVs and FCEVs have been on the rise. The extant literature contains studies on both standalone and grid-connected RE systems for EVs. Podder et al. (2022) supplied the energy demands of an EV charging station in Bangladesh with solar and biogas. The winning system has a levelized cost of energy (LCOE), initial cost, net present cost (NPC), and payback period of $0.181/kWh, $93,530, $19,735, and 12 years, respectively. 100 plug-in hybrid EVs were simulated by Schetinger et al. (2020) based on three scenarios (grid only, RE only (wind/solar), and RE with option of vehicle to grid). The third scenario presented the most optimal results among the three options. The annual operating cost and LCOE of the third scenario are $242,437.26 and $0.12/kWh, respectively. In the work of Minh et al. (2021), the feasibility of EV charging stations running on solar power with and without grid connection was investigated. For the PV-powered charging station, the LCOE for the different investigated locations in Vietnam at a minimum feed-in-tariff of $0.0838/kWh ranges between $0.08-$0.0992/kWh. Techno-economic analysis of solar and wind-powered EV charging stations was investigated by Al Wahedi &Bicer (2022). A hydrogen and ammonia fuel cell and biodiesel generator were added to generate additional power for the charging station, which was designed to supply 50 EVs in Qatar. The NPC of the optimal configuration lies between $2.53-$2.92 million with an LCOE of $0.285-$0.329/kWh. A solar-wind EV charging station was investigated for Turkey by Ekren et al. (2021). The most feasible system produces 843,150 kWh/yr, of which 44.4% and 55.6% are generated by wind turbines and solar PV, respectively, and the cost of producing 1 kWh at $0.064.

Other researchers have also made progress toward refuelling stations for FCEVs. Chen et al. (2021) designed a refuelling station in China for 50 FCEVs with a tank capacity of 5 kg each. The various supply routes analyzed produced an LCOE of 0.28–0.83 RMB/kWh and an LCOH of about 40–50 RMB/kg. The application of wind and solar energy for hydrogen refuelling stations in Turkey has been investigated Gökçek &Kale (2018). The analysis includes 25 FCEVs at a tank size of 5kg/vehicle. The daily hydrogen requirement was estimated at 125 kg. The LCOH of the hybrid solar and wind system was $8.92 against $11.08 of the wind-only system. The former recorded an LCOE and NPC of $1.90/kWh and $8.4 million compared to the latter's $2.50/kWh and $11.1 million, respectively. A similar work (wind only) has been completed for South Africa by Ayodele et al. (2021) for 25 FCEVs (tank size 5 kg/EV) with a yearly hydrogen requirement of 45.63 tons per year. The NPC of the system lies between $3.74-$5.38 million, LCOE ($0.098-$0.140/kWh), and LCOH ($6.34–8.97/kg). A wind-only-powered hydrogen refuelling station in Sweden has also been investigated by Siyal et al. (2015). The investigation has been conducted based on the daily requirement of 200 FCEVs with tank capacities of 8 kg. According to the researchers, an increase in wind speed from 4.5 to 5 m/s can lead to a decrease in LCOH by at least 17%. A hydrogen refuelling station powered by solar power is investigated in the work of Micena et al. (2020). By replacing 10%–100% conventional Taxis with hydrogen fuel, the associated hydrogen requirement per day lies between 20 and 185 kg; LCOE lies between $0.27–0.41/kWh; LCOH ranges from $8.96-$13.55/kg.

Table 1 summarizes the aim, location, power sources, connection type, and fuel demand of the studies reviewed supra. The rationale behind the choices is mostly according to the discretion of the authors of those studies. However, investigating the need for electric or refuelling charging stations could be influenced by future transport policy scenarios in the given case study area; deciding the type of energy resource to investigate could be influenced by the renewable energy availability and potential of the particular case study area; connection type could be influenced by the reliability, installed capacity and excess electricity production of the existing grid network of the study area; finally, the energy demand per day depends on the sample EVs being considered by taking the product of the number of EVs considered and their battery capacity (BEV) or tank capacity (FCEV).

4. Methodology

The methodology adopted for realizing the objectives of the current study is reported in this section. The HRECS is modelled for the fuel demands of potential EVs in West Africa, and Ghana has been selected as the case study. Due to similarities in climatic conditions, demographic, energy resources, and requirements in the region, the method can successfully model HRECS in other countries within the region.

4.1. HOMER software

Several models such as HOMER, iHOGA, PVsyst, SAM, RETScreen Expert, EnergyPlan, etc., have been developed over the years to simulate, optimize, and assess the performance of renewable energy systems. In terms of hourly assessment tools or energy systems, HOMER arguably surpasses the rest as having the best balance of simplicity and flexibility (Thirunavukkarasu &Sawle, 2022). Some major advantages of HOMER over other tools include (i) medium-simplicity, (ii) coherent and integrated programming structure, (iii) capability to model several supply technologies including renewable, non-renewable, and different types energy storages (iv) The ability to simulate electric, thermal, and hydrogen loads enables engineering and economics to be used in tandem and to provide insightful information (Cuesta et al., 2020; Elkadeem et al., 2021; Lyden et al., 2018). Whether designing hybrid microgrids or distributed generation systems, the HOMER software solutions: Combine engineering and economics in one powerful model; Allow users to quickly and efficiently determine least-cost options; Simulate real-world performance and deliver a choice of optimized design; Consider geographic-specific criteria and risks, such as fuel price changes, load growth, accelerated battery degradation and changing weather patterns (HOMER Energy, 2022). In off-grid or grid-tied systems, HOMER determines the optimal size of the components by taken into consideration the following six types of input data: (a) meteorological data (wind speed, solar radiation, temperature, biomass

Table 1
Related studies of standalone and/or grid-tied EV charging/refuelling stations.

Country	Aim	Power source	Connection type	Energy demand/day	Reference
Bangladesh	Electricity charging station	Solar and biogas	Standalone	72.51 kWh	Podder et al. (2022)
Brazil	Electricity charging station	Wind, solar, grid	Standalone and grid-tied	900 kWh	Schetinger et al. (2020)
Vietnam	Electricity charging station	Solar, grid	Grid-tied, grid only, and standalone	–	Minh et al. (2021)
Qatar	Electricity charging station	Solar, biodiesel, and wind	Standalone	1750	(Al Wahedi and Bicer, 2022)
Turkey	Electricity charging station	Solar and wind	Standalone	2.4 MWh	Ekren et al. (2021)
China	Hydrogen refuelling station	Solar, wind, grid	Standalone and grid-tied	500 kg	Chen et al. (2021)
Turkey	Hydrogen refuelling station	Solar and wind	Standalone	125 kg	Gökçek &Kale (2018)
South Africa	Hydrogen refuelling station	Wind	Standalone	125 kg	Ayodele et al. (2021)
Sweden	Hydrogen refuelling station	Wind	Standalone	1600 kg	Siyal et al. (2015)
Brazil	Hydrogen refuelling station	Solar	Standalone	20–185 kg	Micena et al. (2020)

availability, hydro, and steam flow data), (b) hourly electric, hydrogen, and thermal demand profile data, (c) economic cost data (operation and maintenance cost, capital cost, replacement cost, fuel price, transaction electricity price, interest rate, lifetime, system fixed capital cost, maintenance cost, and emissions penalty), (d) technical data (dispatch strategy and operating reserve), (e) equipment characteristics data, and (f) HOMER optimizer or search space (Bahramara et al., 2016; Kim et al., 2017). Based on the specified constraints to the input data, HOMER examines all feasible configurations as potential hybrid systems for adoption, and ranks them according to their net present cost from lowest to highest.

4.2. Profile of case study (Accra)

The focus of the current study is on Ghana's capital city, Accra. Located in the southern part of Ghana, the latitude and longitude of Accra are 5°33′21″ N and 0°11′48″ W, respectively, and 33 m above sea level. Accra covers a land area of 225.67 km^2 with an estimated population of 2.1 million as of 2014 (Miezah et al., 2015). Fig. 2 is the map of the case study area within Ghana. In the present work, primary RES, including biomass, wind energy, and solar power, have been selected due to their huge potential for electricity generation in the country. Referring to NASA's resource website, solar radiation, temperature, and wind speed of the selected location at 5°36.2′N latitude and 0°11.2′W longitude were obtained. Section 4.3 briefly highlights the resource assessment of the selected location for HRECS in Ghana.

4.3. Resource assessment

4.3.1. Solar energy

The annual average solar radiation of the case study area is 4.94 kWh/m^2/day, which represents a monthly average of a 22-year period data collected by NASA. As represented in Fig. 3, the peak periods of solar radiation typically occur from November to April, whiles relatively lower levels of solar radiation are observed from May to October - within this latter period is the rainy season in Southern Ghana. The peak solar radiation occurs in March (5.140 kWh/m^2/day), whiles the lowest solar radiation occurs in June (4.430 kWh/m^2/day). The location also experiences an annual average clearness index of 0.497, and the highest and lowest months for relatively clearer atmosphere are in January and September, respectively.

4.3.2. Wind energy

The wind speed of the selected location was obtained from NASA surface meteorology and solar energy database. The data has been collected over a ten-year period at 50 m above the earth's surface. The annual average wind speed of the area is 3.67 m/s, and the highest and lowest windy periods are in July–September and May–June (together with December), respectively (Fig. 4).

4.3.3. Biomass

Utilizing biomass for electricity generation does not only reduce dependence on fossil-based electricity but also reduces the amount of untreated waste, eliminates environmental burdens of poor waste management, and improves community development through the provision of jobs and zero waste societies (Afrane et al., 2021). The biomass

Fig. 2. Location of Accra in the Southern part of Ghana (Aryee et al., 2018).

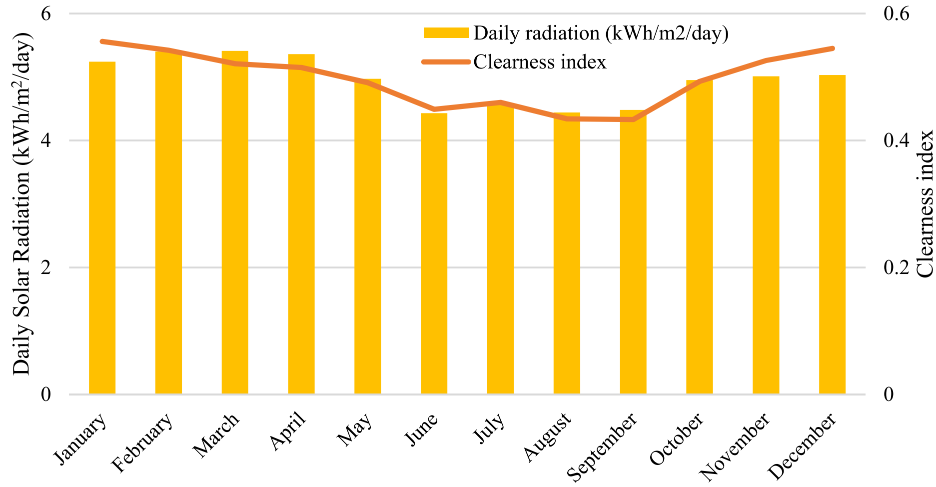

Fig. 3. Seasonal solar resource characteristics of selected location for HRECS (NASA, 2022b)

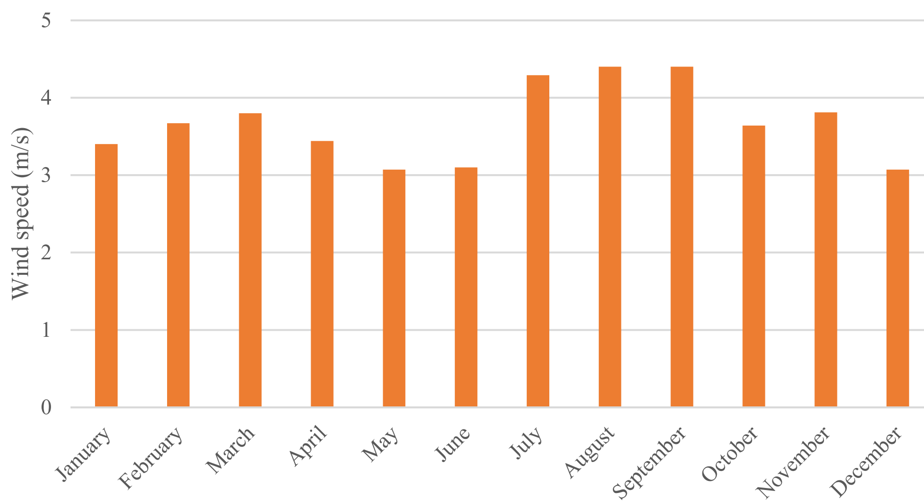

Fig. 4. Seasonal wind resource characteristics of selected location for HRECS (NASA, 2022b).

resources available in the selected area include agricultural waste, MSW, and sewage sludge, which could be collected and harnessed for electricity production. The current study considers MSW as the source of bioenergy for HRECS. In Accra, the MSW generation is estimated at 0.74 kg/p/day (Miezah et al., 2015). Considering the population of Accra, according to Miezah et al. (2015), about 1500 tonnes of MSW is

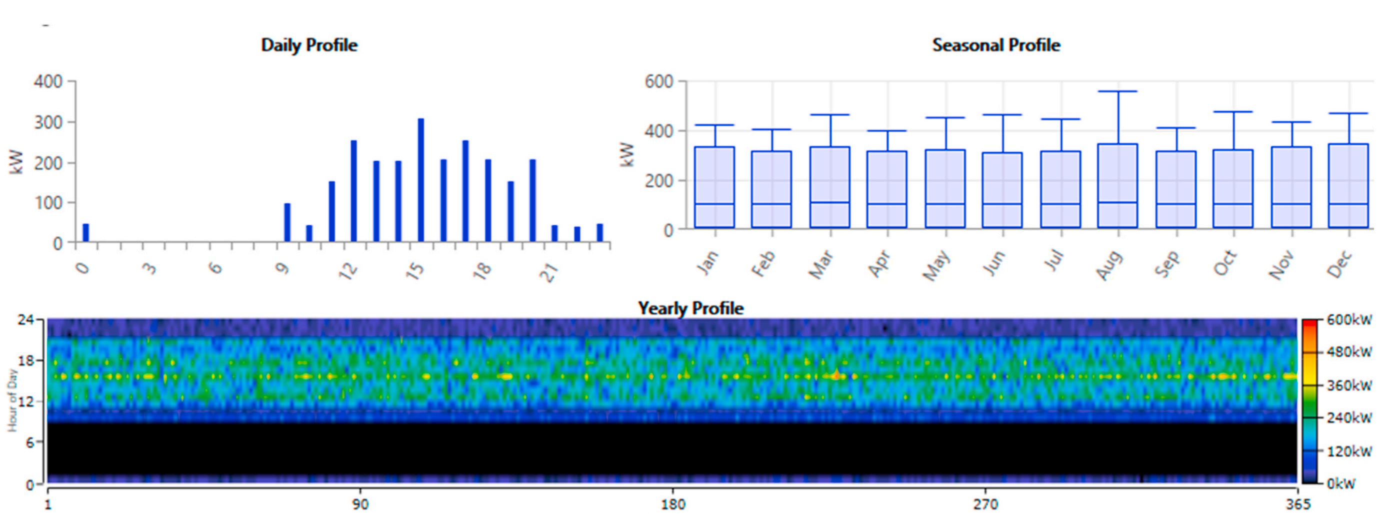

Fig. 5. Load profile of charging station for 70 BEVs.

generated per day. The amount of MSW available for gasification alongside its properties is discussed in subsequent sections.

4.4. Load demand

4.4.1. BEV demand

The electricity demand for the BEV charging station is estimated in this section. The current analysis has been conducted for a potential 100 EVs expected in Ghana, of which 70% are assumed to be BEVs. Considering a battery capacity of 35 kWh, the daily electricity demand of these 70 BEVs is calculated as 2450 kWh. The peak load and average load of the charging station are 556.45 kW and 102.08 kW, respectively, resulting in a load factor of 0.18. The scaled annual value is at a day-to-day and timestep random variability of 10% and 20%, respectively. The load profile of the charging station (Fig. 5) has been modelled based on that of Al Wahedi &Bicer (2022). The expected peak periods of charging occur between the hours of 12:00–21:00.

4.4.2. FCEV demand

The hydrogen demand for the refuelling station for FCEVs follows similar steps as that of the BEVs. 30 FCEVs are considered in the current analysis, resulting in a scaled annual average of 150 kg/day, assuming the tank size of the EVs is 5 kg each. The peak and average flow rates of the refuelling station are 21.3 kg/h and 6.26 kg/h, respectively, with a load factor of 0.29. The scaled annual value is at a day-to-day and timestep random variability of 10% and 20%, respectively. The load profile of the refuelling station (Fig. 6) is modelled based on that of Chen et al. (2021). The expected peak periods of refuelling occur between the hours of 9:00–20:00.

4.5. Operating strategy, inputs, and constraints

HOMER operates on a system of principles termed a 'dispatch strategy' that controls the operations of the storage device(s) and generator(s). Two types of dispatch strategies can be modelled in HOMER: cycling charging (CC) and load following (LF). CC is an optimal strategy in designs with relatively lower or no renewable power; here, the strategy allows the generator to run at full capacity to meet the load and charges the battery bank with surplus electricity. On the other hand, when the project has higher renewable power, the LF strategy is suitable as it only draws the required power from the generator to supplement that of the renewable power in serving the load. LF is a reliable strategy, with higher renewable energy penetration and relatively lower emissions. Accordingly, the LF strategy was selected for the current study. The logic of the operation of the LF strategy is shown in Fig. 7. For this strategy, the amount of power generated by the RES (wind and solar) is compared against the total energy demand of the HRECS. If the supply exceeds the load requirement, the system battery is charged to its maximum limit with the excess power, and any other remaining power after this point is dumped. However, if the power supply from RES is interrupted, the unserved load is served by the battery if the amount of stored electricity is sufficient. Otherwise (inadequate stored electricity in batteries), the remaining part of the load which has not been served will be served by the Biogas generator.

In HOMER, constraints are provided that instruct the optimization process to discard any configuration that goes contrary to the stated constraints. These configurations are unfeasible and are therefore not included in the evaluation of HOMER. In the current study, three unique constraints are considered: a capacity shortage constraint of 0% is set for this study, implying that any configuration that does not meet approximately 100% of yearly energy demand and operational reserve should be treated as not feasible, hence, discarded. The second constraint targets the operating reserve, which adds excess operational capacity to the system. This provides reliability and resilience to the system in the event of an increase in demand or decrease in renewable energy supply. Under this constraint, four inputs are used by HOMER in calculating the needed operational reserve: load in the current time step, annual peak load, and two percentages of renewable energy generation. In the current work, these values were set at 10%, 0%, 80% (solar power output), and 50% (wind power output), respectively. An unmet hydrogen load of 0% was specified, indicating that any system that does not meet 100% of the hydrogen load throughout the year should be considered unfeasible by HOMER and hence discarded.

4.6. Technical assessment and components' cost

4.6.1. Solar PV

Solar cells, also called photovoltaic (PV) cells, convert sunlight directly into electricity. During the day, when sufficient solar radiation is available, these panels are expected to produce power, but their operation is significantly curtailed during the night, on cloudy or rainy days. During such an intermittent supply of solar power, other sources of energy in the design are expected to compensate for the demands of the two loads. Eqn. (1) is used to calculate the output of the PV array (P_{PV}) (HOMER Pro, 2018d);

$$P_{PV} = Y_{PV} f_{PV} \left(\frac{\overline{G}_T}{\overline{G}_{T,STC}} \right) \left[1 + \alpha_p \left(T_c - T_{c,STC} \right) \right] \tag{1}$$

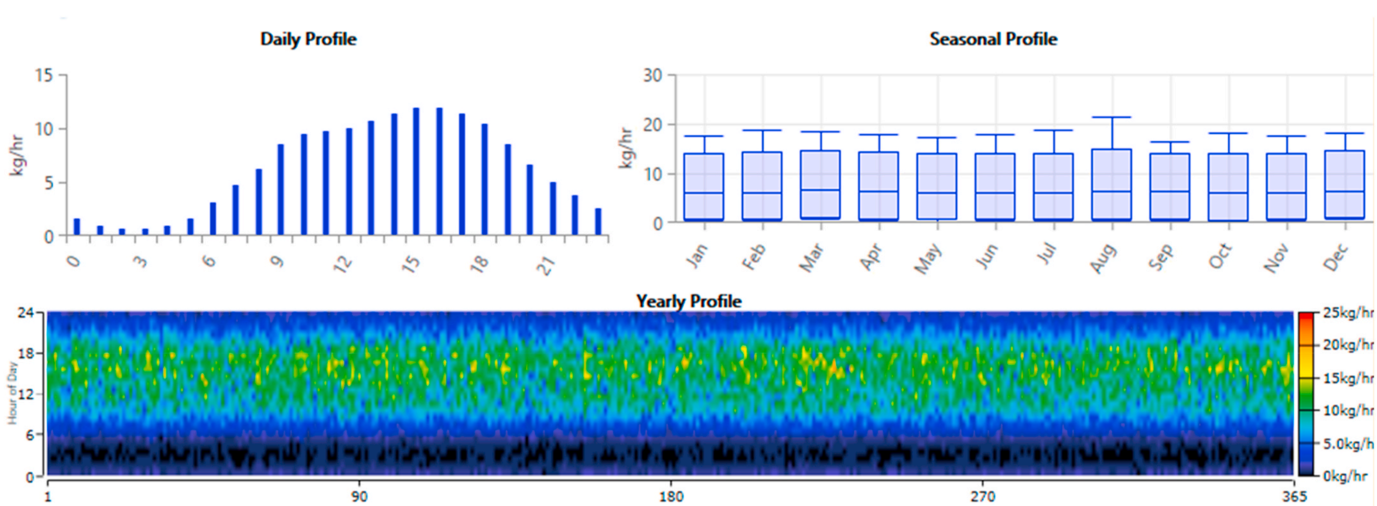

Fig. 6. Load profile of refuelling station for 30 FCEVs.

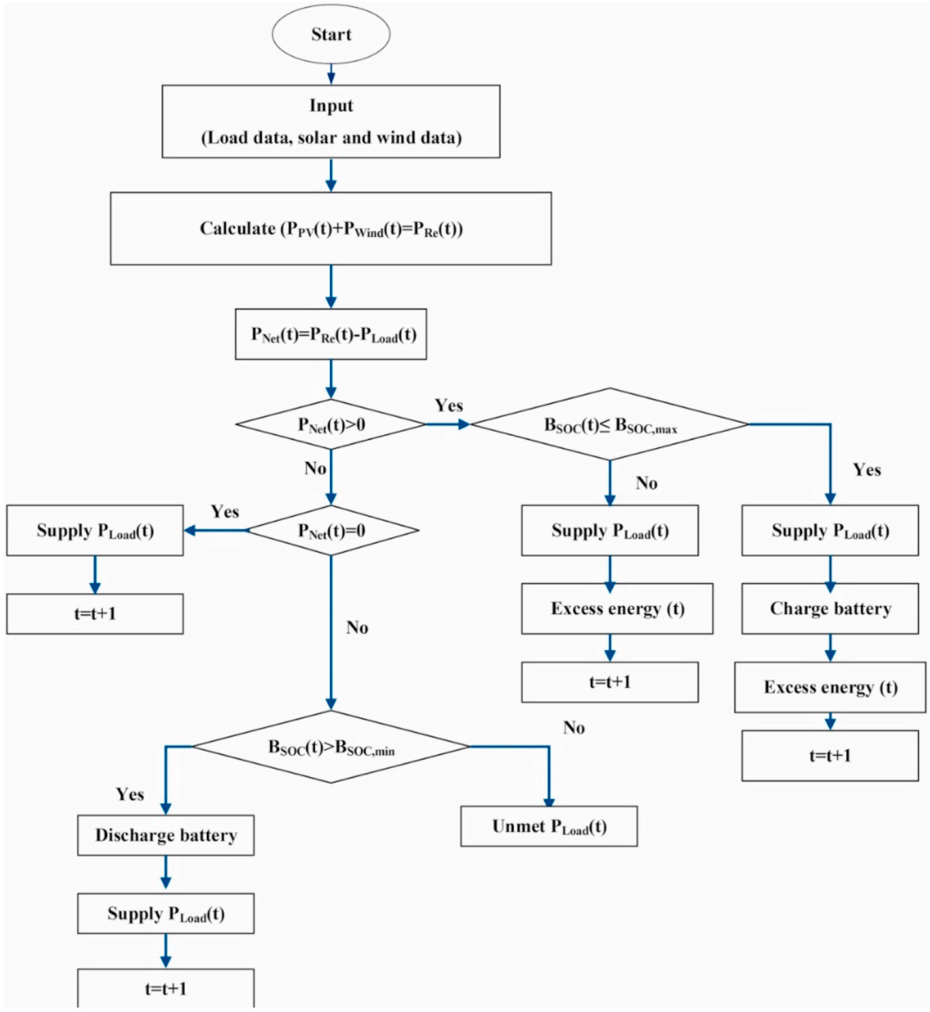

Fig. 7. Logical operation of the load following strategy (Das et al., 2021).

Where – Y_{PV} – rated capacity of PV array under standard test conditions (kW); f_{PV} - PV derating factor (%); $\overline{G}_T$- solar radiation incident on the PV array in the current time step (kW/m^2); $\overline{G}_{T,STC}$- incident radiation at standard conditions (1 kW/m^2); α_p – temperature co-efficient of power (%/°C); T_c- PV cell temperature in the current timestep (°C); $T_{c,STC}$- PV cell temperature under standard test condition (25 °C). Without including the effect of temperature on the PV array, Eq. (1) is simplified into Eqn. (2);

$$P_{PV} = Y_{PV} f_{PV}\left(\frac{\overline{G}_T}{\overline{G}_{T,STC}}\right) \tag{2}$$

4.6.2. Wind turbine

The energy in the wind is harnessed by a wind turbine to convert this energy into mechanical energy, and with the aid of an electrical generator, the mechanical energy is also converted into electricity. In the proposed system design, a wind turbine supplements the solar PVs during the night or on overcast days. The wind turbine power output is calculated using Eqn. (3) (HOMER Pro, 2018c);

$$P_{WTG} = \left(\frac{\rho}{\rho_0}\right)\cdot P_{WTG,STP} \tag{3}$$

Where, P_{WTG}- wind turbine power output (kW); $P_{WTG,STP}$- wind turbine power output at standard temperature and pressure (kW); ρ - actual air density (kgm^{-3}); ρ_0- air density at standard temperature and pressure (1.225 kgm^{-3}).

4.6.3. Biogas generator

In HOMER, the term 'biogas' refers to gasified Biogas. The product can be called one of several different names, including synthesis gas, syngas, producer gas, or wood gas. The role of the Biogas generator (BG) is to generate supplemental electricity from the available MSW. During low or no outputs of solar power, especially during the night or on cloudy days (or Scenarios without PV), the Biogas generator, together with the wind turbine and battery system, is supposed to compensate for the unserved load. In a scenario of no PV or wind turbine, the BG should perform this function alone with the batteries. The BG used in the present work has been scheduled to operate under optimized mode 24/7 throughout the entire year. $40/tonne has been considered as the average cost of the MSW (Al-Najjar et al., 2022). As already mentioned, the MSW produced at the selected area is 1500 t/d. In Ghana, about only 65% of solid waste is collected (Mudu et al., 2021). Also, the country's solid waste is made up of ~66% organic fraction and ~20% combustible fraction (Miezah et al., 2015). Similar estimation has been done in the work of Dodo et al., (2022). Given this information, the amount of daily combustible waste available for power generation by gasification is 195 tonnes (Fig. 8). The carbon content of MSW and gasification ratio are taken as 47.6% (Lee et al., 2020) and 0.70 kg gas/kg MSW (AlMohamadi, 2021; Luo et al., 2012), respectively. The lower heating value (LHV) of the resulting biogas (syngas in reality) is also taken as 20 MJ/kg (Ibikunle et al., 2019). The BG is connected to an AC output with a minimum load ratio set at 20% lifetime operating hours of 9125 h.

Fig. 8. Monthly biomass availability for gasification.

4.6.4. *Storage battery*

Since renewable energies are characterized by reliability issues and are significantly dependent on the climatic conditions of an area, backup systems like storage batteries are strongly recommended in these configurations. Providing this storage unit ensures reliability and a constant supply of voltage to load. The battery receives its power from the excess electricity produced from the other power-generating systems. Surrette 4 KS 25 P battery was used for the analysis. The lifetime of each battery was 20 years.

4.6.5. *System converter*

A converter is integrated into the hybrid system to maintain the flow of energy between AC electrical load and DC components of the hybrid energy system. The efficiency of the rectifier and inverter was kept at 90% and 95%, respectively.

4.6.6. *Electrolyzer and hydrogen tank*

An electrolyzer's role in the setup is to consume electrical energy to split water into hydrogen and oxygen through electrolysis. The lifetime was taken as 15 years. A polymer electrolyte membrane (PEM) electrolyzer with an efficiency of 85% was considered for the current work. The electrolyzer's rate of hydrogen production is estimated using Eqn. (5) (Rad et al., 2020).

$$q_{H_2} = \eta_f \frac{n_c I_{EI}}{2F} \tag{5}$$

Where the rate of hydrogen is denoted by q_{H_2}, the number of cells in series is represented by n_c, η_f denote the Faraday efficiency which can be estimated using Eqn. (6) (Rad et al., 2020), the Faraday coefficient is represented by F and the electrolyzer current is also denoted by I_{EI}.

$$\eta_f = 96\exp\left(\frac{0.09}{I_{EI}} - \frac{75.5}{I_{EI}^2}\right) \tag{6}$$

After the electrolyzer produces the needed hydrogen gas, a storage tank is used to collect and store the fuel which can be used to meet the hydrogen load demand of refuelling station at the mining site.

4.6.7. *Compressor, cooling unit, dispenser*

It is worth noting that the hydrogen produced by a conventional electrolyzer is at a much lower pressure (~14 bar) than the required pressure (350 or 700 bar) at the refuelling station. To realistically model a hydrogen refuelling station, a compressor would be needed to obtain the necessary pressure at a minimum of 300 bar (Gökçek &Kale, 2018). Eqn. (7) is used to determine the actual power (kW) of the compressor (Chen et al., 2021):

$$W_{comp} = (z)\,(R)\,(n)\frac{T_1}{\eta_c}\left(\frac{r}{r-1}\right)\left[\left(\frac{P_2}{P_1}\right)^{\frac{r-1}{nr}} - 1\right] m_c \tag{7}$$

Where the mean compressibility factor of hydrogen is denoted as z, and can typically lie between 1 and 1.4 at 150 K–300 K and 20 bar–350 bar (Gustafson, 2013). The inlet gas temperature of the hydrogen compressor (T_1) was taken as 293 K, η_c which is the compressor efficiency was considered 0.75 in the current study. n is the number of compression stages which has been calculated as 3.9 in the current analysis (details to calculation of n can be referred from Khan et al. (2021) pg. 14). P_1 and P_2 denote the hydrogen compressor's inlet and output gas pressures and are taken as 20 and 350 bar, respectively, and the isentropic exponent, r, of hydrogen is usually considered as 1.4. m_c represents the gas flow rate through the hydrogen compressor measured in kg/s taken as 0.01 kg/s (33 kg/h) (NREL, 2014). The theoretical energy to compress hydrogen isothermally from 20 bar to 350 bar (5000 psi or ~35 MPa) is 1.05 kWh/kg H_2 (DOE, 2009). At the outlet of the electrolyzer and compressor, a low-pressure and high-pressure storage tank is installed, respectively. The peak-hour hydrogen demand and throughput of the compressor are used to determine the size of the high pressure storage tank (Reddi et al., 2016). The ratio of the calculated actual power to the motor efficiency (95% in this study) of the compressor gives its rated power capacity (Khan et al., 2021).

A dispenser with 350 or 700 bar nozzles transfers hydrogen into the tank of hydrogen vehicles from the high-pressure tank (Ayodele et al., 2021). To compensate for the temperature increase introduced as a result of the expansion of the vehicle's tank, a pre-cooling unit is needed. Type B70 hydrogen dispenser was considered in our analysis due to its compliance with the SAE TIR J2601 worldwide hydrogen refuelling protocol. This protocol indicates that an optimal dispenser should have the capability to dispense hydrogen at a pre-cooled temperature of −20 $^\circ C$. In our analysis, the energy consumption required to cool a kg of hydrogen from 15 to −20 $^\circ C$ was considered as 0.18 kWh – a value previously considered in the works of (Ayodele et al., 2021; Gökçek &Kale, 2018). These peripheral components' load demand was modelled as an AC electrical load in HOMER Pro. The lifetime of cooling unit and dispenser were taken as 15 years, and that of dispenser and compressor as 10 years.

Table 2 summarizes the lifetime cost of the HRECS components.

Table 2
Lifetime cost of HRECS components.

Component	Capital cost	Replacement cost	O&M costs	Reference
PV	$350/kW	$350/kW	$10/kW/yr	Ur Rashid et al. (2022)
Wind turbine	$1000/kW	$1000/kW	$50/kW/yr	Ayodele et al. (2021)
Biogas generator	$200/kW	$200/kW	$0.03/kW/yr	(Al Wahedi and Bicer, 2022)
Electrolyzer	$100/kW	$100/kW	$8/kW/yr	Ur Rashid et al. (2022)
Low pressure hydrogen storage tank	$600/kg	$400/kg	$3/kg	Siyal et al. (2015)
Storage battery	$200/unit	$200/unit	$3/unit/yr	Agyekum &Nutakor (2020)
System converter	$300/kW	$300/kW	–	HOMER Pro (2018b)
High pressure hydrogen storage tank	$1448/kg	$1014/kg	$3/kg/yr	Gökçek &Kale (2018)
Cooling unit	$5000/unit	$5000/unit	–	Gökçek &Kale (2018)
Dispenser	$54,000/unit	$54,000/unit	–	Gökçek &Kale (2018)
Compressor	$2500/kW	$2500/kW	–	Ayodele et al. (2021)

4.7. Economic assessment

4.7.1. Net present cost

The net present cost (NPC) of the system is calculated by taking into account the difference in the present value of all outgoing cash flows (expense) and that of all incoming cash flows (revenue) throughout the lifetime of the project. NPC is calculated using Eqn. (8) (Gökçek &Kale, 2018);

$$C_{npc,tot} = \frac{C_{ann,tot}}{CRF} \tag{8}$$

Where the total annualized cost ($/year) is denoted by $C_{ann,tot}$ and the capacity recovery factor is represented by CRF. Eqn. (9) is used to calculate the CRF of the project (Gökçek &Kale, 2018);

$$CRF = \frac{i(1+i)^n}{(1+i)^n - 1} \tag{9}$$

Where i is the real discount rate, and n is the project lifetime.

4.7.2. Levelized cost of energy

The levelized cost of electricity (LCOE) is the cost involved in producing 1 kWh of electricity. Eqn. (10) is used to calculate the LCOE of standalone and grid-connected hybrid systems (Jia et al., 2020);

$$LCOE = \frac{C_{ann,tot}}{E_{served}} \tag{10}$$

Where, E_{served} is the sum of the primary load consumption and sales to grid in kWh/year. For standalone systems, the sale to grid term is zero.

4.7.3. Levelized cost of hydrogen

Levelized cost of hydrogen is the cost involved in producing 1 kg of hydrogen. It is determined by Eqn. (11) (HOMER Pro, 2018a);

$$LCOH = \frac{C_{ann,tot} - V_{elec}\left(E_{prim,AC} + E_{prim,DC} + E_{def} + E_{grid,sales}\right)}{M_{hydrogen}} \tag{11}$$

Where, V_{elec} is the selling price of the electricity that will be sold and injected into the electrical grid ($/kWh); E_{prim} is the primary electrical load (kWh/year), E_{def} is the deferrable load (kWh/year), $E_{grid,sales}$ is the total energy sold to the grid (kWh/year), and $M_{hydrogen}$ is the total hydrogen production (kg/year). For the current work, Eqn. (12) reduces to Eqn. (12);

$$LCOH = \frac{C_{ann,tot} - V_{elec}\left(E_{prim} + E_{grid,sales}\right)}{M_{hydrogen}} \tag{12}$$

In standalone systems, the term V_{elec} is zero since no electricity is injected from the excess electricity into the grid. Hence, in these systems, the LCOH is as shown in Eq. (13);

$$LCOH = \frac{C_{ann,tot}}{M_{hydrogen}} \tag{13}$$

4.7.4. Payback period

The payback period refers to the number of years taken for one to recover the initial investment of a project. Beyond this period, the investor begins to accrue profit on their investment. A project with a payback period beyond the project's lifetime is considered not feasible and vice versa. The lower the payback period, the more attractive and feasible a project becomes. The payback period of a proposed project can be calculated using Eqn. (14);

$$Payback\ period = \frac{Initial\ capital}{Annual\ savings} \tag{14}$$

Where the payback period is in years, initial capital is the capital cost required for establishing the proposed project in $, and the annual inflow of cash/revenue measured in $/year is the annual savings. By using Eqn. (15), the annual savings can be calculated (Manoj Kumar et al., 2019);

$$Annual\ savings = Electricity\ generation\ per\ year \times FiT\ rate \tag{15}$$

Where FiT stands for feed-in-tariff measured in $/kWh. As of 2016, the Public Utilities Regulatory Commission (PURC, Ghana) has provided the FiT of various RES (PURC 2016), which have been adopted for the current analysis. The FiT in Ghana Pesewas per kWh of wind, solar, and biomass are 65.3529, 59.7750, and 69.1225, respectively, which were converted to US$ before applying them in Eqn. (15).

4.7.5. Profitability index

The profitability index (PI) is an economic indicator for ascertaining the attractiveness of a project from an economic point of view. The higher the PI, the more attractive the project becomes. It is defined as the ratio of cash inflow to cash outflows expected once the project begins. If PI is less than 1, the project is not worth embarking on, and vice versa. PI is calculated using Eqn. (16) (Karmaker et al., 2020);

$$PI = \frac{Cash\ inflow\ per\ year \times Project\ lifetime}{Total\ investment} \tag{16}$$

4.8. Environmental assessment

4.8.1. Avoided CO_2 emissions from electricity generation

Power generation from fossil-based fuels such as coal, natural gas, and oil leads to excessive release of GHGs, especially CO_2, into the atmosphere. It is, therefore, necessary to consider the CO_2 emissions of any project related to electricity production and consumption. RES such as solar and wind provide eco-friendly means of electricity generation because the production does not lead to the emissions of GHGs. Therefore, the amount of CO_2 mitigated by powering HRECS with RES instead of fossil-based electricity can be calculated using Eqn. (17) (Podder et al., 2022);

$$Mit_{CO_2} = E_g \times F_e \tag{17}$$

Where, Mit_{CO_2} are the mitigating CO_2 emissions, E_g represents the annual electricity production of the RES for the HRECS system, and F_e is the emission factor taken as 0.35 $kgCO_2/kWh$ (Nyasapoh et al., 2022) in the current study.

4.8.2. Gasoline replacement potential and avoided CO_2 emissions

By using Eq. (18) (Ayodele et al., 2021), the amount of gasoline fuel replaced with hydrogen in FCEVs is calculated;

$$M_{GFuel} = \frac{M_{H_2} \times LHV_{H_2}}{LHV_{GFuel}} \tag{18}$$

Where M_{GFUEL} is the potential replacement of gasoline in kg, M_{H2} is the amount of hydrogen produced on-site per year, LHV_{H2} and LHV_{GFUEL} are the lower heating values of hydrogen and gasoline fuel taken as 120 MJ/kg and 44 MJ/kg, respectively (Onorati et al., 2022). Similarly, the potential CO_2 emissions avoided during the replacement of gasoline with hydrogen are calculated using Eq. (19)

$$A_{CO_2} = M_{GFuel} \times SE_{CO_2} \tag{19}$$

Where A_{CO2} is the amount of CO_2 emissions avoided per year in kg; M_{GFUEL} is the mass of the replaced gasoline in kg, and SE_{CO2} is the specific emission factor of CO_2, taken as 2.3 kg per 1 kg of gasoline (Ayodele et al., 2021).

5. Multicriteria decision assessment

As mentioned in section 1, the selection of the most feasible renewable energy system for electricity production and hydrogen generation is not a straightforward task, especially when considering several multicriteria factors related to technical, economic, and environmental feasibilities. In the current work, six different systems (Scenario A-F) are considered (Fig. 9). All factors described in section 4 are kept constant with the exception of the power source discussed herein: Scenario A consists of a Biogas generator only; Scenario B consists of a solar PV only; Scenario C consists of a Biogas generator and wind turbine; Scenario D consists of a solar PV and wind turbine; Scenario E consists of a Biogas generator and solar PV; Scenario F consists of a Biogas generator, solar PV and wind turbine. Due to the relatively lower average wind speed in the selected area, a wind turbine-only system is not considered.

The sizing of the components of each Scenario can be seen in the attached supplementary file. The sizing has been carefully done in such a manner that the minimum size at which the system can produce feasible results is considered. This ensures unbiased and accuracy as high as possible. The optimal size of HRECS's components for each scenario after the HOMER simulation has been made available in the supplementary file. Overall, 14 evaluation criteria are included in the assessment, including five technical (electricity production, hydrogen generation, excess electricity, unmet load, capacity shortage), five economic (net present cost, levelized cost of energy, levelized cost of hydrogen, initial capital, payback period), and four environmental factors (CO_2 avoided from electricity generation, gasoline replacement potential, CO_2 avoided from gasoline replacement, GHG emissions). Fig. 10 shows the hierarchical framework for the selection of the most feasible energy system for HRECS in Ghana.

Based on the objectives of the current study, MCDA models CRITIC and PROMETHEE II are selected for the analysis, and the steps for both models are described below.

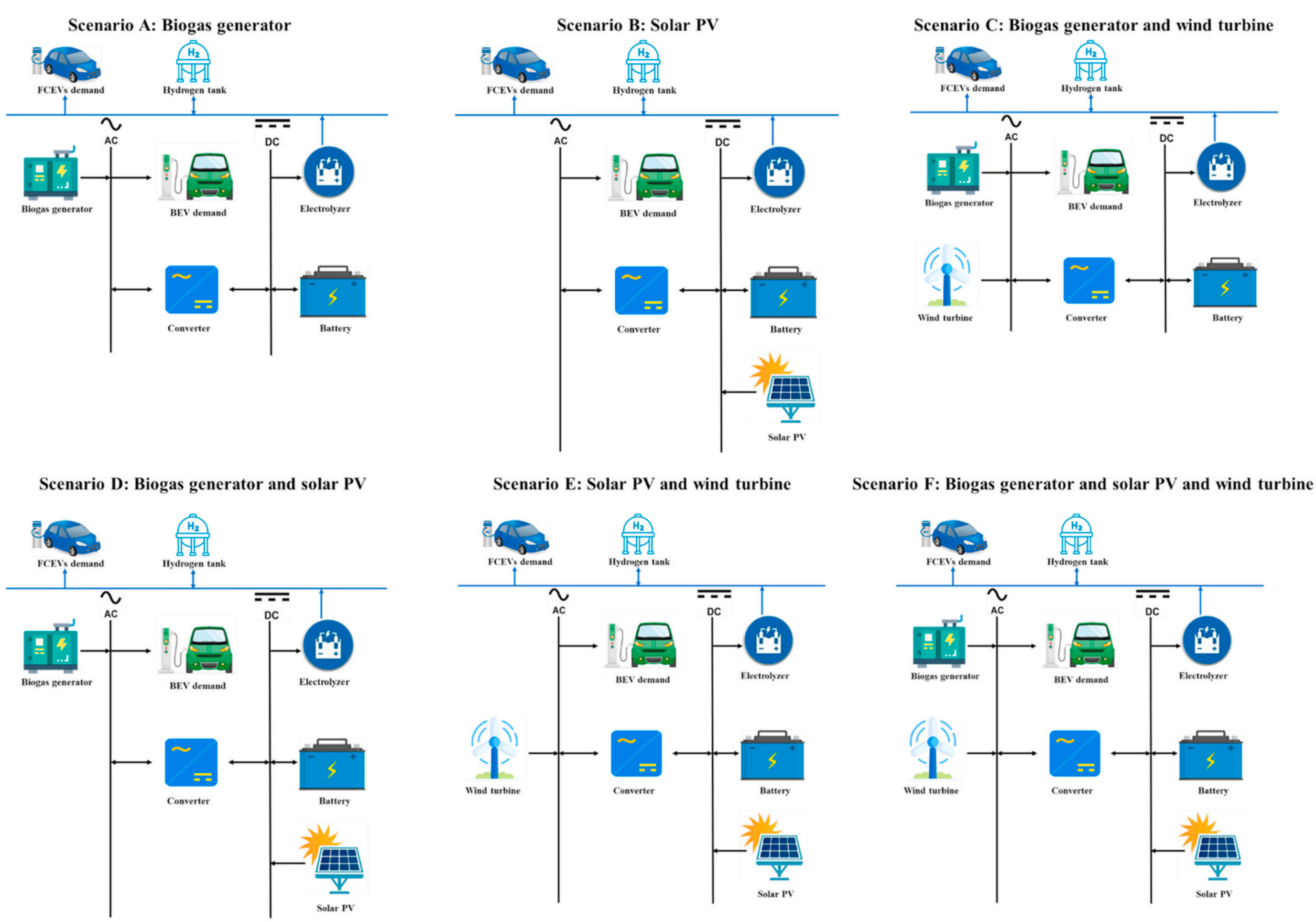

Fig. 9. Potential RE systems for HRECS.

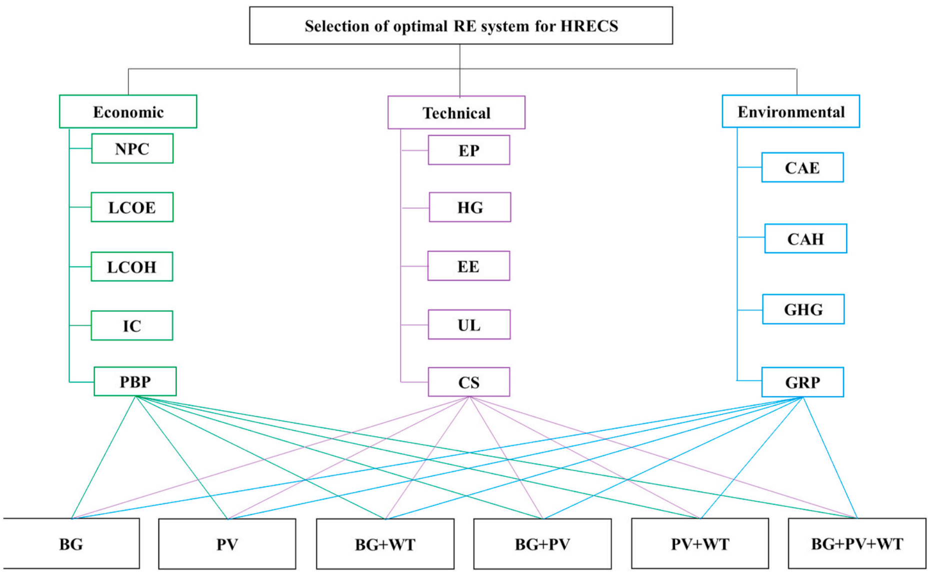

Fig. 10. Hierarchical framework for prioritizing the most feasible setup for HRECS in Ghana. Note – NPC: net present cost; LCOE: levelized cost of electricity; LCOH: levelized cost of hydrogen; IC: initial capital; PBP: payback period: EP: electricity production; HG: hydrogen generation; UL: unmet load; EE: excess electricity; CS: capacity shortage; GHG: greenhouse gas emissions; CAE: carbon dioxide avoided from electricity generation; CAH: carbon dioxide avoided from gasoline replacement with hydrogen; GRP: gasoline replacement potential; BG: Biogas generator; PV: solar photovoltaic; WT: wind turbine.

5.1. CRITIC

The basic characteristic of the CRITIC is the intensity of the constant (Nguyen et al., 2020). The objective weights calculation is built as follows (Bhadra et al., 2022; Kumaran, 2021; Rani et al., 2021):

Step 1Using Eqn. (20), a normalized decision matrix is obtained from the performance scores of each scenario;

$$r_{ij} = \frac{x_{ij} - x_j^{min}}{x_j^{max} - x_j^{min}}; i = 1, \ldots, m; j = 1, \ldots, n \tag{20}$$

Where, x_{ij} represents the performance scores of scenarios, x_j^{min} and x_j^{min} denote the worst and best value amongst the scenarios under each criterion, respectively.

Step 2Based on the obtained normalized decision matrix, the standard deviation σ_j of the jth criterion is calculated
Step 3A symmetric matrix of n × n with element r_{jk} which refers to the linear corelation coefficient between any 2 criteria is determined. X is calculated using Eqn. (21);

$$r_{jk} = \frac{\sum_{i=1}^{m}\left(r_{ij} - \bar{r}_j\right)\left(r_{ik} - \bar{r}_k\right)}{\sqrt{\sum_{i=1}^{m}\left(r_{ij} - \bar{r}_j\right)^2 \sum_{i=1}^{m}\left(r_{ik} - \bar{r}_k\right)^2}}; k = 1, \ldots, n \tag{21}$$

Step 4Each criterion's measure of conflict created by the jth criterion with respect to the decision situation designed by the rest of criteria is calculated using Eqn. (22);

$$Conflict = \sum_{k=1}^{n}\left(1 - r_{jk}\right) \tag{22}$$

Step 5. By multiplying the standard deviation in step 2 to the measure of conflict in step 4, the quality of information C_j contained in the jth criterion is calculated using Eqn. (23);

$$c_j = \sigma_j \times \sum_{k=1}^{n}\left(1 - r_{jk}\right) \tag{23}$$

Step 6. Compute the result of objective weights using Eqn. (24);

$$W_j = \frac{c_j}{\sum_{k=1}^{n} c_j} \tag{24}$$

As a result, the objective weights of all selection criteria are determined when the above steps are executed. The objective weights are accepted when they rank from 0 to 1 (Diakoulaki et al., 1995).

5.2. PROMETHEE II

The steps related to the PROMETHEE II ranking of the hybrid RE systems are outlined as follows (Athawale &Chakraborty, 2010; Debbarma et al., 2017; Paul et al., 2015):

Step 1Determine the weighted normalized decision matrix:

$$R_{ij}\left\{x_{ij} - min\left(x_{ij}\right)\right\} / \left\{max\left(x_{ij}\right) - min\left(x_{ij}\right)\right\} \tag{25}$$

(i = 1,2,3 … …n: j = 1, 2, 3 … … m).Where x_{ij} is the preference measure of the *ith* hybrid RE system on *jth* evaluation criteria. It should be noted that Eqn. (25) is for beneficial criteria; For non-beneficial criteria, Eqn. (26) is used as follows

$$R_{ij}\left\{max\left(x_{ij}\right) - \left(x_{ij}\right)\right\} / \left\{max\left(x_{ij}\right) - min\left(x_{ij}\right)\right\} \tag{26}$$

Step 2The differences of the *ith* hybrid RE system relative to the other hybrid RE system is calculated. This stage includes calculating the differences in criteria values between various paired systems.
Step 3Calculation of $P_j(i, i')$. The six different forms of preference functions necessitate the establishment of specific preferential criteria, which is indifference thresholds and preference. The following equation is used to determine the appropriate preference function of each criterion due to real-time application challenges, especially when decisions were made:

$$P_j(i, i') = 0 \; if \; R_{ij} \leq R_{i'j} \tag{27}$$

$$P_j(i, i') = \left(R_{ij} - R_{i'j}\right) \; if \; R_{ij} > R_{i'j} \tag{28}$$

Step 4The aggregated preference function is determined using criteria weights, as shown below:

$$\Pi\left(i, i'\right) = \left[\sum_{j=1}^{m} W_j \, X \, P_j\left(i, i'\right)\right] \Big/ \sum_{j=1}^{m} W_j \tag{29}$$

i.e. w_j represents the relative importance (weight) of *jth* criteria.

Step 5 The entering (negative) and leaving (positive) outranking flows are determined using:

Leaving flow for *ith* hybrid RE system,

$$\vartheta^{+} = \frac{1}{(n-1)}\left\{\sum_{i=1}^{n}\pi(i,i^{'})\right\}(i \neq i^{'}) \tag{30}$$

Entering flow for *ith* hybrid RE system,

$$\vartheta - = \frac{i}{(n-1)}\left\{\sum_{i=1}^{n}\pi(i^{'},i)\right\}(i \neq i^{'}) \tag{31}$$

Where, n represents the number of possible solutions. In this case, each hybrid RE system is up against $(n-1)$ other alternatives. The ϑ^{+} indicates how much one system dominates the other systems, whereas the ϑ^{-} represents how much the other systems dominate a system. The PROMETHEE II technique can yield the complete preorder by employing a net flow based on these outranking flows.

Step 6 The net outranking flow for each of the hybrid RE systems were calculated using:

$$\vartheta(i) = \vartheta^{+}(\mathrm{i}) - \vartheta^{-}(\mathrm{i}) \tag{32}$$

Step 7 This stage based on the values of $\vartheta(i)$, the ranks of all hybrid RE systems were determined. Highly ranked hybrid RE systems will have higher values of $\vartheta(i)$, whiles the poor hybrid RE systems will have a relatively lower values of $\vartheta(i)$.

Fig. 11 shows the conceptual framework adopted in realizing objectives of current study.

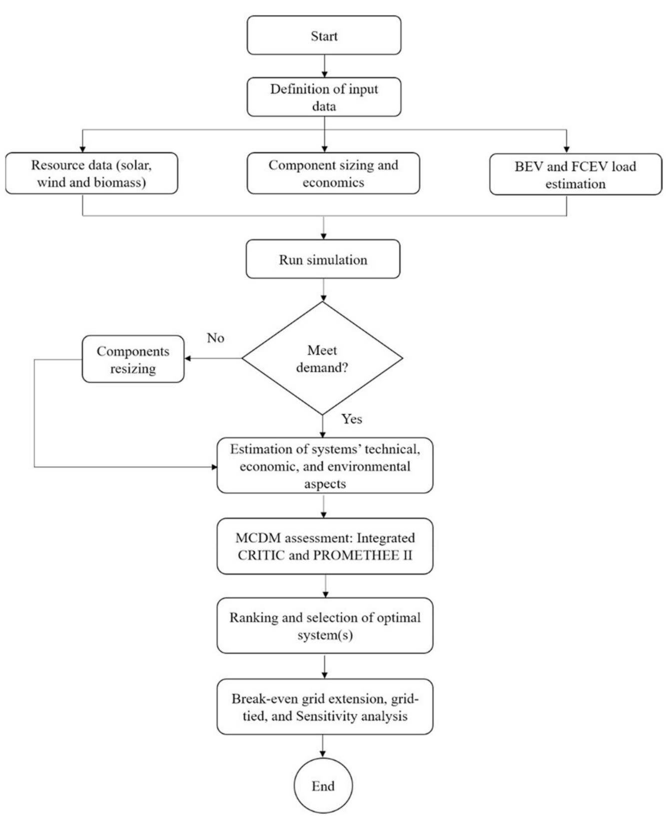

Fig. 11. Conceptual framework adopted in current study.

6. Results and discussion

The characteristics of each Scenario based on their technical, economic, and environmental feasibilities have been reported in this section. Selected results of each Scenario are briefly discussed first before presenting the MCDM results for the most attractive and feasible Scenario among the six alternatives.

Scenario A: Powered by Biogas generator only, the NPC of this Scenario is $22,283,012, produces annual electricity of 3,876,000 kWh, with avoided CO_2 emissions from electricity generation and gasoline replacement of 1,356,600 kg/yr and 344,484 kg/yr, respectively. The cost of producing hydrogen in this Scenario is $27.62/kg.

Scenario B: HRECS in this Scenario is powered by PV only. The cost of electricity generation in this Scenario is $0.33/kWh with hydrogen and electricity production capacity of 55,660 kg/yr and 6,492,097 kWh/yr, respectively. The Scenario results in excess electricity of 43.3%, and the CO_2 emissions avoided from gasoline replacement are calculated as 349,140 kg/yr.

Scenario C: In this Scenario, the source of electricity is a Biogas generator and a wind turbine. The Scenario is characterized by an LCOH of $29.5/kg, gasoline replacement potential of 151,808 kg/yr, a payback period of 11.63 years, and GHG emissions of 20,484 kg/yr. In this scenario, the demand was 100% met at all times.

Scenario D: Biogas generator and PV are the main power-generating systems of Scenario D. The initial capital required to set up this scenario is $2,361,376, and the setup leads to a production of 55,576 kg/yr of hydrogen and 3,892,249 kWh/yr of electricity. The excess electricity available in this scenario is 7.61%, and the total GHG emissions are recorded as 4029 kg/yr.

Scenario E: The electricity supply of this Scenario is by PV and wind turbines. Some of the economic features of this scenario are the payback period (6.49 years) and LCOE ($0.37/kWh); some technical features also include capacity shortage (0.1%) and unmet load (0.03%); whiles the avoided CO_2 emissions from electricity generation, and gasoline replacement is 2,329,716 kg/yr and 349,147 kg/yr, respectively.

Scenario F: The energy demand in this Scenario is from a PV, a Biogas generator, and a wind turbine. Some characteristics of this 3-powerplant system include, LCOE ($0.48/kWh), LCOH ($8.36/kg), electricity production (3,902,118 kWh/yr), excess electricity (7.63%), and gasoline replacement potential (151,475 kg/yr). Complete results of all Scenarios have been provided in Table 3.

Next, the results of CRITIC-PROMETHEE II MCDM are discussed. The objective weight of all 14 criteria based on Table 3 was calculated using the CRITIC model. According to the results, the five most important selection criteria for the proposed project are capacity shortage, excess electricity, unmet load, payback period, and initial capital. The complete results of CRITIC weight estimation are provided in the supplementary material.

By using the calculated objective weights, the six scenarios were ranked by the PROMETHEE II model, and the results are displayed in Fig. 12. Scenario B (PV only) looks promising for the proposed project, especially from an economic and environmental perspective. However, the Scenario is characterized by reliability issues – with an unmet load of 0.04% and capacity shortage of 0.1%, a fraction of the demand is left unserved irrespective of how small it is. This can be attributed to the PV's inability to operate at full capacity throughout the day, especially during the night, dawn, cloudy or rainy days, and the electricity available from the storage batteries is not enough to meet the unserved load during these periods. Furthermore, the scenario has a huge fraction of excess electricity (43.3%), which is unutilized and overly exceeds the demand if it were to be reserved for dump loads. Thus, though promising, the CRITIC-PROMETHEE II model ranked Scenario B in third place. The 3-power-generating setup in Scenario F also presents some interesting findings. The energy demand is 100% met with no capacity shortage and an acceptable level of excess electricity (7.63%). The MCDM model ranks this Scenario in second place, and it will serve as the

Table 3
Feasibility of six different Scenarios for HRECS.

Scenarios	Scenario A	Scenario B	Scenario C	Scenario D	Scenario E	Scenario F
Economic results						
NPC ($)	22,283,012	4,121,526	21,231,602	6,528,248	4,600,872	5,999,379
LCOE ($/kWh)	1.79	0.33	1.71	0.52	0.37	0.48
LCOH ($/kg)	31.38	5.72	29.5	9.09	6.39	8.36
IC ($)	1,911,376	2,851,376	3,921,376	2,361,376	3,341,376	3,411,376
PBP (yr)	5.47	5.69	11.63	7.65	6.49	11.09
Technical results						
EP (kWh/yr)	3,876,000	6,492,097	3,778,705	3,892,249	6,656,332	3,902,118
HG (kg/yr)	54,918	55,660	55,663	55,576	55,661	55,541
UL (%)	0	0.04	0	0	0.03	0
EE (%)	4.69	43.3	1.75	7.61	45.0	7.63
CS (%)	0	0.1	0	0	0.1	0
Environmental results						
GHG (kg/yr)	24,162	0	20,484	4029	0	2042
CAE (kg/yr)	1,356,600	2,272,234	1,322,547	1,362,287	2,329,716	1,365,741
CAH (kg/yr)	344,484	349,140	349,158	348,613	349,147	348,393
GRP (kg/yr)	149,776	151,800	151,808	151,571	151,803	151,475

Note – NPC: net present cost; LCOE: levelized cost of electricity; LCOH: levelized cost of hydrogen; IC: initial capital; PBP: payback period: EP: electricity production; HG: hydrogen generation; UL: unmet load; EE: excess electricity; CS: capacity shortage; GHG: greenhouse gas emissions; CAE: carbon dioxide avoided from electricity generation; CAH: carbon dioxide avoided from gasoline replacement with hydrogen; GRP: gasoline replacement potential.

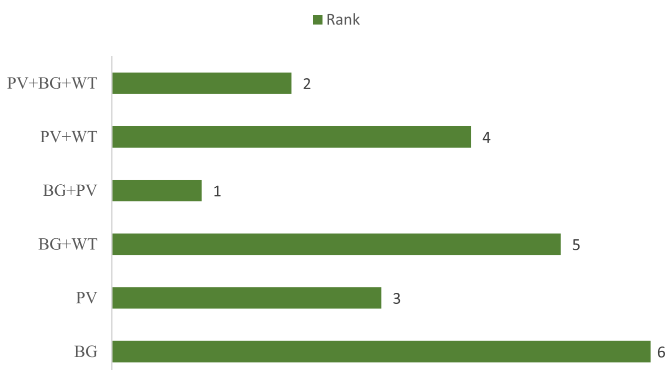

Fig. 12. Final rank of six potential scenarios for HRECS in Ghana.

baseline scenario against the winning scenario for comparative purposes in subsequent sections. Similar to Scenario F, Scenario D (Biogas generator and PV) served all available loads at all times with no capacity shortage and an acceptable level of excess electricity (7.61%). However, compared to Scenario F, Scenario D has a better initial capital, payback period, hydrogen production capacity, higher gasoline replacement potential, and slightly lower excess electricity. These are 5 out of the available 14 selection criteria, and the adopted hybrid MCDM model ranked Scenario D at first place ahead of Scenario F. The final results of the CRITIC-PROMETHEE II ranking have been made available in the supplementary material.

In subsequent sections, a critical analysis is given of the technical, economic, and environmental aspects of Scenario D. All other related analyses of the current study will revolve around this particular scenario.

6.1. Technical feasibility

The system architecture of the winning scenario comprises a 2 MW generic solar PV (SM500), 1.5 MW generic Biogas generator, a 7.55 kWh capacity battery (79 strings), a 432 kW system converter, and a 1.5 MW and 1000 kg electrolyzer and hydrogen tank, respectively. The technical feasibility of Scenario D begins with the electricity production per year. The system generates 3.9 GWh of electricity annually for HRECS, of which 83.4% and 16.6% are generated by the solar PV and Biogas generator, respectively. The electricity production of the winning scenario is about 0.26% lower than that of the baseline scenario. Fig. 13 shows the monthly average electricity production of Scenario D. The production is high from October to May and gradually reduces from June to August. This could be attributed to the fact that the rainy season in the selected location occurs during the latter period and could thus explain the relative drop in electricity production from the PV.

The AC primary load, in this case, the BEV demand and the additional components (compressor and cooling unit) consume 961,775 kWh/yr of electricity (24.7% of the total electricity produced). Fig. S1 shows the hourly BEV load served for each month within the year. The load-serving is particularly spontaneous between 9:00 to 20:00, when relatively significant electricity is demanded to charge the BEVs. No load was served from the hours of 1:00 to 9:00 as no BEV is expected for charging in this period.

The rated capacity of the PV is 2 MW with a mean output of 8893 kWh/day and a capacity factor of 18.5%. The penetration rate of the PV is 338%, and it operated for 4380 h/yr. Fig. 14 shows the PV output for each day in the year on an hourly basis. It is seen that from 18:00 to 7:00, the PV output is virtually 0 as there is no or insignificant amount of solar radiation during this period to produce any useful amount of power. The demand during this low or no PV output is expected to be met by the

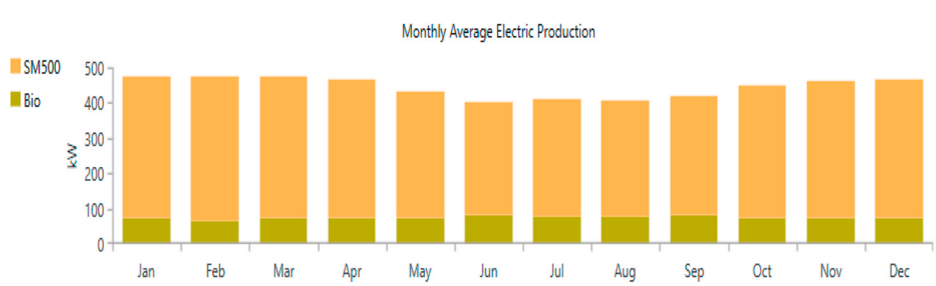

Fig. 13. Monthly average electricity production by Scenario D.

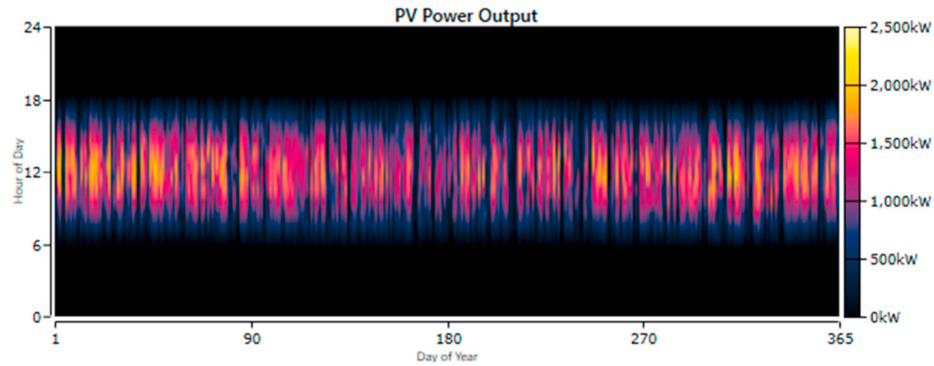

Fig. 14. Yearly PV power output.

Biogas generator and the storage batteries.

The rated capacity of the Biogas generator is 1.5 MW. It consumes 2308 tons of the available MSW annually and has a specific fuel consumption of 2.50 kg/kWh. Throughout the year, the generator observed 763 starts and produced 646,200 kWh/yr of electricity for HRECS. Fig. S2 shows the Biogas generator's power output per year. It is seen that during the day (when there is sunshine), the generator supplemented some of the electricity produced by the PV to serve the load. Also, during hours between 15:00 to 22:00, when there was little or no PV output, the Biogas generator provided relatively significant amount of the required electricity. The power output of the generator, however, drops after 22:00 to 1:00, but virtually no output from 1:00–11:00. The drop in biogas generator output during periods of low/no PV output could be attributed to the fact that the load requirement in those periods were relatively low or non-existent or partially/completely served by the storage batteries.

79 batteries of 4 V each connected in parallel, have been used to serve the load during hours of low or no PV and Biogas generator output. The batteries have a useable nominal capacity of 358 kWh, and a lifetime throughput of 833,584 kWh. The batteries stored 110,709 kWh of electricity and discharged 88,760 kWh each year. The storage depletion and losses were recorded as 215 kWh/yr and 22,165 kWh/yr, respectively. Fig. 15 shows the state of charge of the batteries throughout the year. It is seen that between 7:00 to 18:00, the state of charge of the batteries was almost maximum, which could be attributed to the high PV output available for serving the load and charging the batteries. However, as the output of the PV diminishes, the state of charge of the batteries also gradually reduces as the batteries rather discharge to serve the available load as much as it can serve.

Scenario D produces 55,576 kg/yr of hydrogen, which is only 0.06% more than that of the baseline scenario. The electrolyzer is rated at 1.5 MW and has a mean input of 294 kW with a capacity factor of 19.6%. The electrolyzer consumes 2,579,025 kWh/yr of the total electricity produced. For every kg of hydrogen produced, 46.4 kWh of electricity was consumed, and the average and peak amount of hydrogen produced by the electrolyzer is 6.34 kg/h and 32.3 kg/h, respectively. Fig. S3 shows the electrolyzer input power for each day across the entire year. It is seen that during low or no output of PV, relatively lower power was available for the electrolyzer. The results imply that the electrolyzer was highly dependent on the available electricity generated by the PV.

The hydrogen load consumed 54,808 kg of the hydrogen produced annually, and the profile of the stored hydrogen and the served FCEV refuelling station is shown in Fig. 16. It is seen that in months were PV output was relatively lower, the amount of hydrogen stored was reduced as the electrolyzer's output reduces and the hydrogen tank discharges more to serve the hydrogen load. Although the PV output is high in January, hydrogen production had just begun and hence relatively lower amount will be available for storage at the early stages of the year. The hydrogen load served was practically consistent throughout the year. When the hydrogen in the tank is below the maximum limit, and there is excess energy available, the electrolyzer will have to produce more hydrogen until the level reaches the set limit (1000 kg), after which the electrolyzer is turned off. However, if there is not sufficient excess electricity available, the electrolyzer is kept off, and the hydrogen tank discharges to meet the hydrogen load until the minimum threshold of the tank is attained. In such situations, an unmet hydrogen load is recorded, and this becomes a loss of hydrogen supplied (Al-Buraiki &Al-Sharafi, 2022). In the current work, the hydrogen tank of Scenario D has a tank autonomy of 304 h, and the hydrogen content at the beginning and at the end of the year was recorded as 100 kg and 868 kg, respectively.

Scenario D, recorded 0% unmet load and with zero capacity shortage. However, the excess electricity of the baseline scenario (7.63%) was 0.26% higher than Scenario D (7.61%). Reducing the size or capacity of the power generating systems to decrease the excess electricity is not an advisable strategy to pursue as this could pose reliability issues to HRECS. Alternatively, the excess electricity could be diverted for a fraction or all dump loads such as heating, cooling, ventilation, and lighting (Gökçek &Kale, 2018). Furthermore, when connected to the grid, the excess electricity can be sold back to the grid to increase the economic feasibility of HRECS. The energy balance of Scenario D has been provided in Fig. 17. It is seen that out of the 3.9 GWh of electricity made available for utilization per year, 98.6% was made useful whilst the remaining was lost as waste. This insignificant loss could be attributed to the efficiencies of the components, charging and discharging of batteries, as well as the instability of equipment

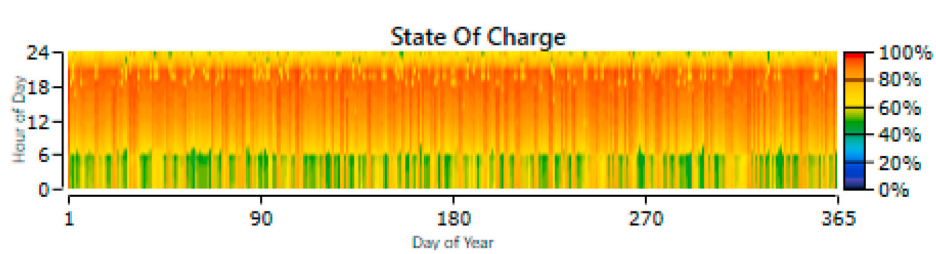

Fig. 15. State of charge of batteries of Scenario D.

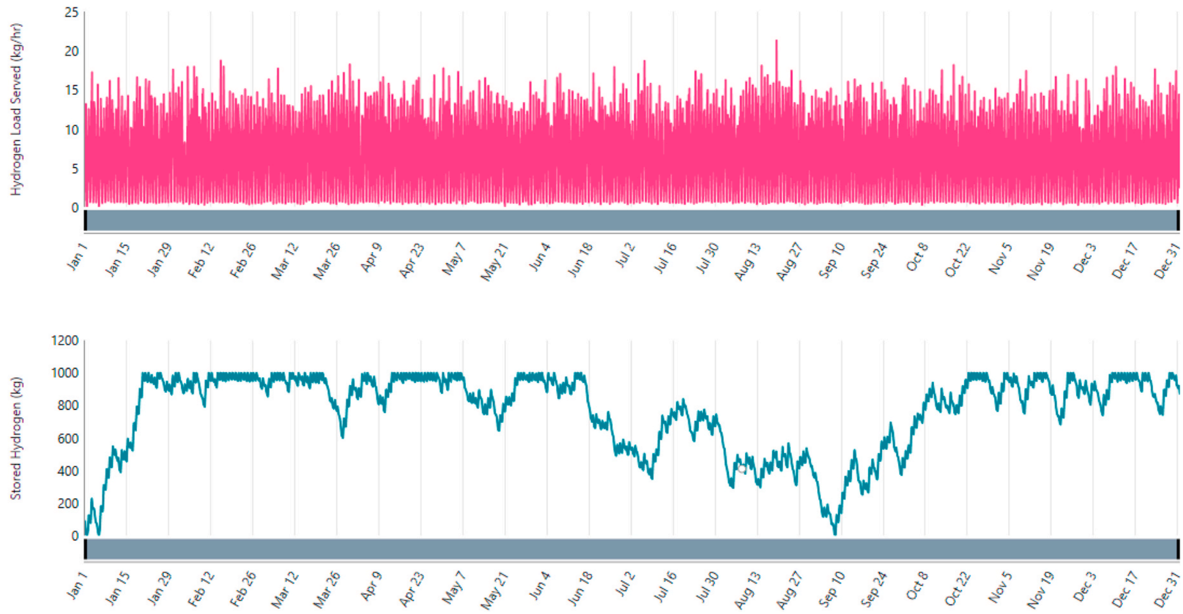

Fig. 16. Hydrogen stored and load served in Scenario D.

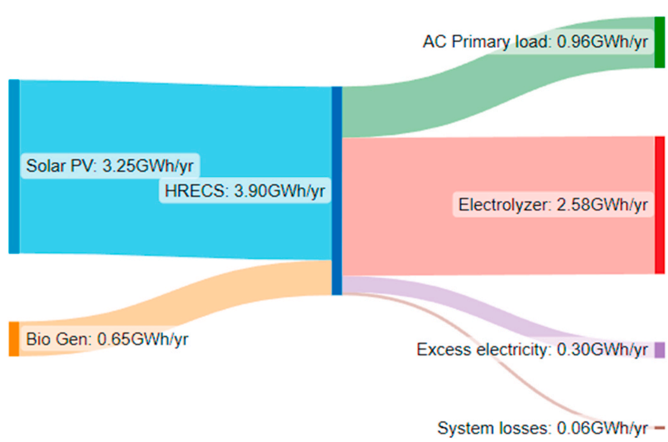

Fig. 17. Energy balance of Scenario D.

operation and maintenance (Jia et al., 2020).

6.2. Economic feasibility

Economic assessment is arguably the most important assessment for RE investments; it enables potential investors to decide whether a particular project is worth investing in. In the current study, the NPC, initial capital, LCOE, LCOH, payback period, and profitability index are the main indicators for assessing the economic feasibility of HRECS.

The proposed project would require at least \$2,361,376 of initial capital to begin operations successfully. Each year, an annual operation cost of \$292,059 is incurred. The NPC of HRECS is \$6,528,248, and the results reveal that at least 80% of this cost is borne by the capital cost (36%), O&M costs (26%), and replacement cost (20%) of the components. In Fig. 18, the distribution of the NPC amongst the HRECS components is shown. It can be observed that despite its relatively smaller contribution to power generation (16.6%) compared to the solar PV (83.4%), a large share of the system's NPC is concentrated in the life cycle cost of the Biogas generator. This is followed by the solar PV, hydrogen tank, electrolyzer, system converter, and batteries, in that order. The high NPC of the Biogas generator can be attributed mainly to its O&M costs and cost of fuel consumption (resources). Compared to the baseline scenario, the NPC of scenario D is 8.82% higher.

The cost of producing electricity in HRECS for Scenario D is \$0.52/kWh, about 8.33% higher than the baseline scenario. The average cost of end-user electricity in Ghana from 2010 to 2020 is around \$0.13/kWh (Sasu, 2022). This observation presents an economic disadvantage to the proposed project. However, a decrease in the unit cost of fuel

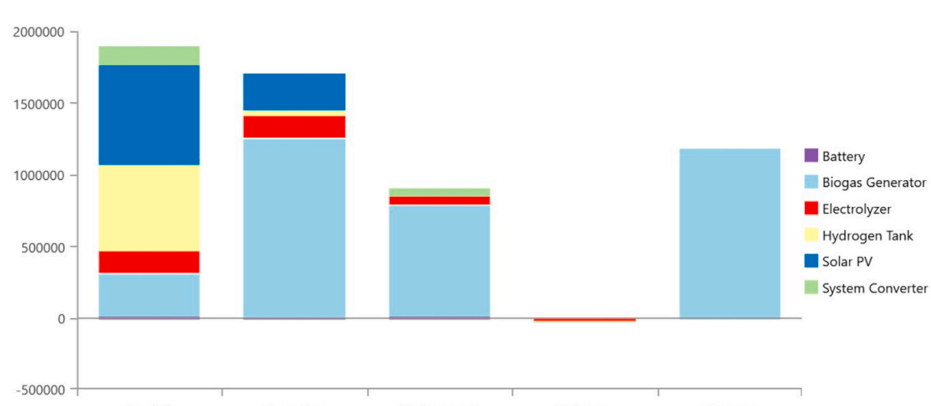

Fig. 18. Share of total HRECS' NPC according to the component type.

consumption and lifecycle cost of components, especially Biogas generator, PV, and hydrogen tanks, in the near foreseeable future could lead to a cheaper generation of electricity than the current situation. Furthermore, the excess electricity of the system could be sold to the grid – which is a potential strategy for reducing the current LCOE of HRECS (Gökçek &Kale, 2018). Fig. S4 shows the share of LCOE amongst the system's components (without the additional components). LCOE is a function of NPC; hence, the trend in Fig. S4 is the same as the reported results in Fig. 18. Biogas generator is responsible for the largest share of HRECS′ LCOE (62%), followed by solar PV (17%), hydrogen tank (11%), electrolyzer (6%), system converter (3%), and batteries (1%).

Previous studies on EV charging/refuelling have reported varying LCOEs; \$1.9-\$2.5/kWh (Gökçek &Kale, 2018), \$0.04-\$0.12/kWh (Chen et al., 2021), \$0.098-\$0.140/kWh (Ayodele et al., 2021), \$0.181/kWh (Podder et al., 2022), \$0.06/kWh (Karmaker et al., 2020), and \$0.285-\$0.329/kWh (Al Wahedi and Bicer, 2022). It should be noted that these studies are for one type of EV station, either BEV charging or FCEV refuelling – hence, the annualized cost of components is relatively lower in these works compared to the current work, which simultaneously provides fuel for both types of EVs at the same station. Besides, in the current analysis, only 27.2% of the electricity production was used in serving the primary electric load (BEV charging + electricity demand of additional components for FCEV refuelling), whiles ~66% was consumed by the electrolyzer for producing hydrogen for the FCEV load. By considering the above-mentioned reasons, in addition to the formula for calculating LCOE, it would make sense why the cost of producing 1 kWh of electricity in the current study is relatively higher compared to previous studies.

Compared to the baseline scenario, the LCOH of Scenario D (\$9.09/kg) is 8.73% higher. According to Megía et al. (2021), the cost of producing 1 kg of hydrogen lies between \$0.054-\$10.36, depending on the production technology. As such, the cost of producing hydrogen at the selected location for HRECS is within acceptable levels. Similar to the case of LCOE, the present LCOH could be further reduced when the cost of fuel consumption and system components reduces. Selling the excess electricity to the national grid also offers an attractive pathway for further reducing the LCOH. The LCOH reported in previous studies include \$6-\$8/kg (Chen et al., 2021), \$8.92-\$11.08/kg (Gökçek &Kale, 2018), \$6.34-\$8.97/kg (Ayodele et al., 2021), \$8.96-\$13.55/kg (Micena et al., 2020), \$5.18-\$9.62/kg (Siyal et al., 2015). As observed, the LCOH of the current study is highly consistent with reported LCOHs for hydrogen refuelling stations in different countries/regions. As mentioned earlier, the electrolyzer consumed 66% of the total available electricity and, in return, generated sufficient hydrogen per year to offset or compensate for the high NPC of the entire system.

For the \$2,361,376 of initial capital invested in HRECS of the current study, it will take 7.65 years for the investment to start accruing profit and a profitability index of 3.27. Previous studies on EVs generate a payback period of 4.99 years (Podder et al., 2022), 14 years (Al Wahedi and Bicer, 2022), and a profitability index of 2.002 (Podder et al., 2022) and 1.2 (Al Wahedi and Bicer, 2022). The results presented in this section reveal that the proposed HRECS has attractive economic characteristics which make it worthy of investing.

6.3. Environmental feasibility

In order to effectively mitigate climate change, the decarbonization of the transport sector cannot be overlooked. About 23% of energy-related CO_2 emissions that cause global warming are emitted by the transport sector. Without an immediate solution, this figure could rise to 40% and 60% by 2030 and 2050, respectively (ITF, 2022). The International Council on Clean Transportation (ICCT) performed an extensive new life cycle assessment (LCA) of GHG emissions from a wide range of typical medium-size passenger car powertrains and fuels. Their analysis revealed that the likely potential pathway to very low GHG emissions in the context of deep decarbonization of the transport sector by 2050 is by a full transition to BEVs and FCEVs only by 2030 (Searle et al., 2021).

Using RE for power generation for electricity required by BEVs and FCEVs (hydrogen from electrolysis) as opposed to fossil-based electricity, usually from grid-powered charging stations, could further boost the environmental aspects of EVs. According to ICCT, the lifecycle GHG emissions from BEV and FCEV powered solely by RE are only 18–22% of GHG emitted by gasoline and diesel cars. Thus, a full transition of conventional passenger cars to EVs is the required scenario needed to avoid the worst impacts of the climate crisis (Searle et al., 2021). Hence, in this section, the environmental-related results for the proposed HRECS project powered by solar and Biogas (Scenario D) are presented.

Due to the carbon content of MSW, the proposed HRECS system emits 4029 kg/yr of GHG emissions, of which 99.8% are contributions from CO_2 emissions. The total GHG emissions of scenario D are 97% higher than that of scenario F. If Ghana adopts carbon pricing policies where environmental burdens of CO_2 emissions are shifted back to the emitters, the current NPC of HRECS will increase by 0.03–0.08% assuming a carbon tax of \$40-\$100/tCO_2 (CPLC, 2017) were to be implemented in the country. The low increase in NPC cost after the implementation of the carbon tax is a result of the high renewable fraction (100%) of Scenario D, which produces relatively lower GHG emissions. By replacing the 1.5 MW Biogas generator with an equal size diesel generator, the renewable fraction of Scenario D reduces from 100 to 33% and produces CO_2 emissions 165 times that emitted in the case of PV + Biogas generator (4020 kg/yr against 664,244 kg/yr). The NPC of this new system (PV + Diesel generator) after a carbon tax of \$40/tCO_2 is at least 45% higher than that of the PV + Biogas generator.

Currently, Ghana 's installed capacity is dominated by thermal power plants; by drawing power from the grid to supply the demand of HRECS, relatively higher emissions are expected as opposed to 100% renewable power. By using Eqn. (17), the CO_2 emissions avoided from generating green electricity are calculated. In a year, 1362 tonnes of CO_2 emissions are mitigated from HRECS. Furthermore, the use of EVs not only presents air pollution benefits but also promotes sustainability by reducing the over-dependence on fossil fuels for transport applications. By replacing 30 gasoline vehicles in Ghana with FCEVs, the gasoline reduction potential (GRP) has been calculated using Eqn. (18) as 152 tonnes/year, and in the process, 349 tonnes/year of CO_2 are avoided (CAH) from the Ghanaian transport sector. In a previous study by Ayodele et al., (2021), for 25 FCEVs in South Africa, the GRP and CAH were obtained as 46.38–46.46 tonnes/year and 106.67–106.86 tonnes/year, respectively. Also, about 4.6 metric tons of CO_2 emissions are emitted by a typical gasoline vehicle per year; since BEVs and FCEVs operate exclusively on electricity and hydrogen, respectively, they do not emit any tailpipe emissions (EPA, 2018). Hence, by replacing 100 gasoline vehicles in Ghana with EVs, at least 460 metric tons of CO_2 emissions are saved annually.

The results obtained in this section reveal relevant environmental benefits of the proposed RE system for HRECS in addition to the already discussed techno-economic feasibility.

6.4. Break-even distance analysis

In this section, the standalone system (Scenario D) is re-simulated, but this time with the option of grid extension. The objective is to compare the economic attractiveness of a standalone system against the extension of the grid. The economic criterion considered for this analysis is the net present cost. The capital cost of grid extension, its operating and maintenance cost, and grid power price are considered as \$8000/km, \$1800/yr/km (Li et al., 2022), and \$0.13/kWh (Sasu, 2022), respectively. In order to reach this result, the Break-even grid extension distance (BGED) analysis is used. At BGED, the NPC of the grid extension is equal to that of the standalone system. BGED is calculated using Eqn. (33) (HOMER Pro, 2018b);

$$D_{grid} = \frac{C_{NPC} \times CRF(i, R_{proj}) - c_{power} \times E_{demand}}{c_{cap} \times CRF(i, R_{proj}) + c_{om}} \tag{33}$$

Where, C_{NPC} refers to the total NPC of the standalone system ($); $CRF(i, R\_proj)$ refers to the capital recovery factor; i is the real discount rate (%); R_{proj} is the lifetime of the project (yr); E_{demand} refers to the total annual electrical demand (kWh/yr); c_{power} represents the cost of power from the grid ($/kWh); c_{cap} refers to the capital cost of grid extension ($/km); c_{om} denotes the O&M cost of grid extension ($/yr/km).

Fig. 19 shows the BGED of the current study. At a BGED of 130 km, the NPC of grid extension and standalone system intersect. Below this intersection point (130 km), the grid extension is more feasible for HRECS, whiles beyond 130 km, the standalone system is a better alternative for HRECS than grid extension. In other words, as the grid extension distance increases, its NPC increases accordingly, further reducing the feasibility of the grid extension whiles the standalone becomes more attractive. Therefore, based on the current results, during real-life application, the distance between HRECS and the grid network should be carefully considered before deciding whether to draw the required electricity from the national grid or a standalone system. However, should the former be chosen, the associated pressure and negative impact on the grid, as well as the highly fossil nature of Ghana's grid power should not be disregarded during the decision-making.

6.5. Grid-tied versus standalone

As indicated earlier, a full grid-powered HRECS has potential negative impacts on the grid network; however, besides a standalone system, a grid-tied RE system is another strategy for reliably meeting the energy demand of HRECS. In this section, the results of a grid-tied RE system are compared against the standalone system (Scenario D). This is to help ascertain which of the two configurations presents more feasible results for adoption considering technical, economic, and environmental factors. From the MCDM results, it was seen that the PV only (Scenario B) presented some interesting results, but the main setbacks were a 0.04% and 0.10% unmet load and capacity shortage, respectively, as well as relatively huge excess electricity. Hence, adding a Biogas generator (Scenario D) corrected this situation. In the grid-tied analysis, the Biogas generator is replaced with the grid power to provide power, especially during low or no PV output periods. Furthermore, the grid is considered 100% reliable, and so the battery units are removed. Besides, electricity could be sold back to the grid to generate additional revenue for the project. Hence, the comparison done in this section is for a Biogas generator + PV setup (BG + PV) against a grid + PV setup (GD + PV). The capacity of the Biogas generator was used to size the grid purchase capacity, and all other parameters remained constant between the two systems for a fair comparison. The capacity of PV in BG + PV is 2 MW; however, that of GD + PV had to increase to 2.3 MW in order to obtain feasible results amidst the specified constraints in this study.

On the HOMER platform, a scheduled rate structure is selected. The sale and annual purchase capacity were set at 290 kW (which was the maximum capacity that could be sold) and 1500 kW, respectively. An electricity purchasing and selling price of $0.13/kWh and $0.08/kWh were considered, respectively. The system architecture of GD + PV comprises a 2.3 MW solar PV, 1.5 MW grid, 313 kW system converter, 1.5 MW electrolyzer, and 1000 kg hydrogen tank. The system produces 4,104,640 kWh of electricity per year, of which 91% and 9% are from PV and grid purchases, respectively (Fig. S5).

An annual net purchase of 264,527 kWh of electricity was recorded, which is the difference between the electricity purchased (371,684 kWh/yr) and sales (107,157 kWh/yr). The electricity purchases, as seen in Fig. 20, mainly occurred at hours of low or no PV output, and on the other hand, during high PV output during the day, excess electricity was sold back to the grid. Table 4 is a complete comparison of the standalone system (Scenario D – BG + PV) against the grid-tied system (GD + PV).

It is seen from Table 4 that both grid-tied and standalone hybrid RE systems have the potential to meet 100% of the yearly energy requirements of HRECS with 0% unmet load and capacity shortage. However, the results show that the grid-tied option presents relatively superior economic characteristics over its standalone counterpart. For instance, the initial capital, NPC, LCOE, and LCOH of the grid-tied system are 11%, 45%, 50%, and 62% lower than that of the standalone system, respectively. On the other hand, the standalone system also depicts relatively better environmental results than the grid-tied system, including a slightly higher gasoline replacement potential and avoided CO_2 emissions from gasoline replacement. Also, its CO_2 emissions (as a result of significantly higher renewable fraction) and excess electricity were lower than the grid-tied system. With respect to technical feasibility, a grid-tied system is optimal for electricity production and lower system losses whiles hydrogen production is slightly higher with respect to standalone system. The results supra suggest that the decision in selecting grid-tied over standalone system and vice versa for HRECS in a Sub-Saharan country will depend on the most important selection criteria of the intended project.

7. Sensitivity analysis

During its lifetime, it is possible for the project to experience certain uncertainties that may positively or negatively affect its feasibility in the long term. As such, during the planning stages of any major investment such as the proposed HRECS, it is important to factor these uncertainties into the present analysis. A sensitivity analysis is one of such methods to

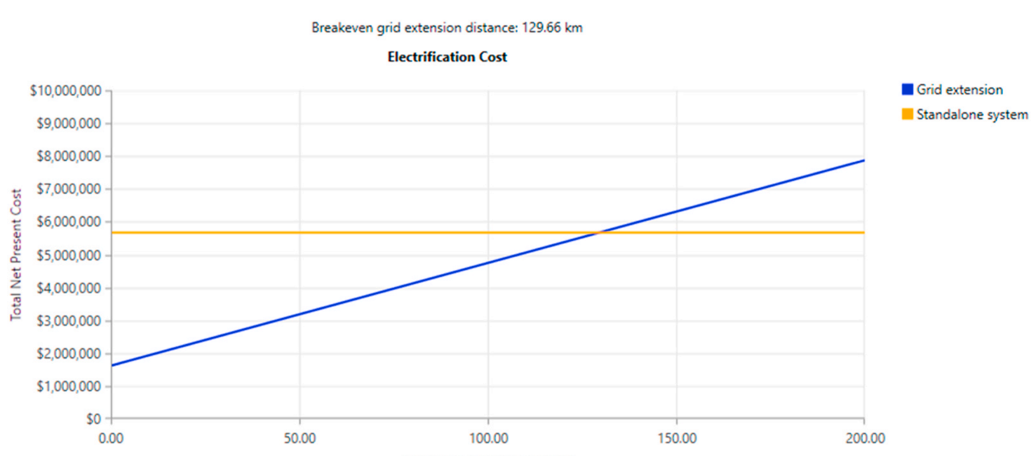

Fig. 19. NPC of standalone versus grid extension (BGED).

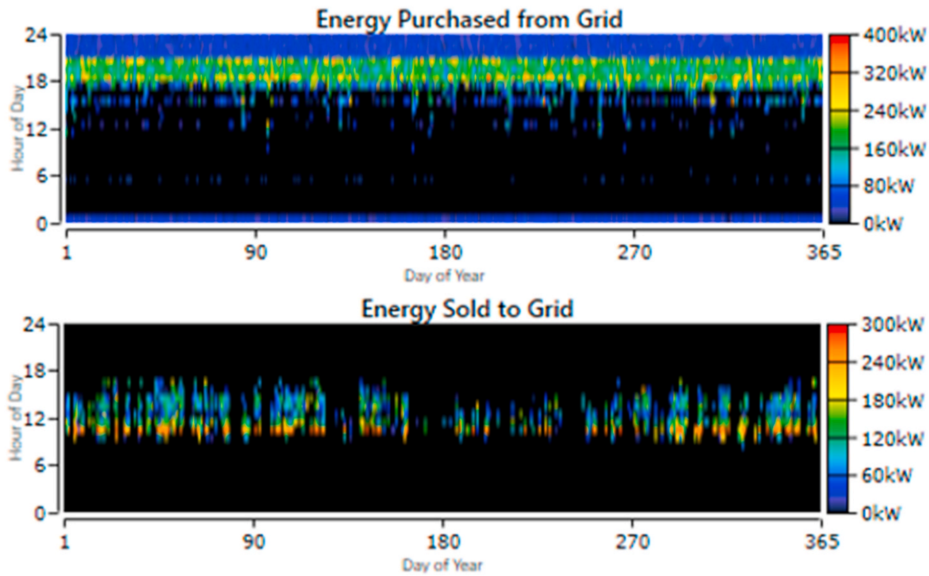

Fig. 20. Electricity transactions between HRECS and grid network.

Table 4
Grid-tied versus standalone for HRECS.

Criteria (unit)	BG + PV	GD + PV
Electricity production (kWh/year)	3,892,249	4,104,640
Excess electricity (%)	7.61	10.3
Unmet load (%)	0	0
Capacity Shortage (%)	0	0
Renewable fraction (%)	100	65.2
Hydrogen production (kg/year)	55,576	55,554
System loss (%)	1.4	0.9
NPC ($)	6,528,248	3,595,178
LCOE ($/kWh)	0.52	0.26
LCOH ($/kg)	9.09	3.47
Initial capital ($)	2,361,376	2,111,376
CO_2 emissions (kg/year)	4020	234,904
Gasoline replacement potential (kg/yr)	151,571	151,511
Avoided CO_2 emissions from gasoline replacement (kg/yr)	348,613	348,475

Table 5
Parameters and their investigated effects on techno-economic feasibility of HRECS.

Parameter (unit)	Variations	Effect investigated
PV cost ratio	1[a], 0.1, 0.3, 0.5, 0.7	NPC and LCOE
Biomass price ($/ton)	0, 20, 40[a], 60, 80	NPC and LCOE
Hydrogen tank cost ratio	1[a], 0.1, 0.3, 0.5, 0.7	LCOH
Minimum load ratio Biogas generator (%)	10, 20[a], 30, 40, 50	Electricity production per year and LCOE
Lifetime of electrolyzer (years)	5, 10, 15[a], 20, 25	LCOH
Efficiency of PV (%)	15, 17.3[a], 19, 22, 24	Electricity production per year and LCOE
Electricity sell back price ($/kWh)	0.05, 0.06, 0.07, 0.08[a], 0.09	NPC and LCOE
Grid sale capacity (kW)	90, 140, 190, 240, 290[a]	NPC and LCOE
Inflation (%)	2[a], 4, 6, 8, 10	NPC and LCOE
Discount rate (%)	2, 4, 6, 8[a], 10	NPC and LCOE

[a] Initial condition.

determine the robustness of the present findings and compare it against future results amidst changes in present conditions. In the current work, the sensitivity analysis is conducted to investigate the technical and economic resilience of the proposed HRECS project in the event of sudden changes in present conditions. The sensitivity analysis also provides a means of obtaining the expected trend in feasibility for similar future projects. Here three main economic (NPC, LCOE, and LCOH) and one technical indicator (electricity production per year) have been analyzed. The parameters varied as shown in Table 5, include PV cost, biomass cost, minimum load ratio of Biogas generator, lifetime of electrolyzer, cost of hydrogen tank, efficiency of PV cells, electricity sell back price, grid sale capacity, expected inflation, and discount rate. It should also be noted that in this part of the analysis, the additional HRECS components (compressor, dispenser, cooling unit, and high-pressure hydrogen tank) has not been included. In other words, only the components on the HOMER platform have been accounted for in the sensitivity analysis. For results involving NPC and LCOE, the LCOE has been superimposed on the NPC whiles the variations in parameters occupy the x and y-axis. For results involving electricity production/year and LCOE, electricity production per year has been superimposed on LCOE.

The effect of the expected inflation rate and nominal discount rate on LCOE and NPC is shown in Fig. 21. Both parameters show an opposite trend on the economic feasibility of HRECS. As the inflation rate increases, LCOE decreases, but NPC increases, whiles at an increasing discount rate, the LCOE increases, but NPC decreases. At an 8% discount rate, when the inflation rate increases from 2 to 10%, the LCOE decreases from $0.414-$0.324/kWh, whiles NPC increases from $4.78-$9.28 million. Also, at a 2% inflation rate, when the discount rate increases from 2 to 10%, the LCOE increases from $0.342-$0.442/kWh, whiles NPC decreases from $7.65-$4.27 million. At a 2% discount rate and 10% inflation rate, the lowest LCOE is achieved, whiles at a 10% discount rate and 2% inflation rate, the lowest NPC is realized. Potential investors of HRECS should, therefore, carefully consider the expected inflation rate and nominal discount rate before deciding on investing.

Characteristics of RE systems such as the minimum load ratio of the Biogas generator or the efficiency of the PV cells have the potential to affect the techno-economic feasibility of the HRECS. The minimum allowable load on the generator is the generator's minimum load ratio, and it is a percentage of its rated capacity. At a specified minimum load, the generator does not have to necessarily shut off; it is simply a mechanism designed to keep a generator from operating at too low a load. Some manufacturers recommend that their generator run above a certain load (HOMER Pro, 2018b). Hence, it is important to investigate how this input affects electricity production and the LCOE of the hybrid RE system. Similarly, researchers in the field are actively focused on

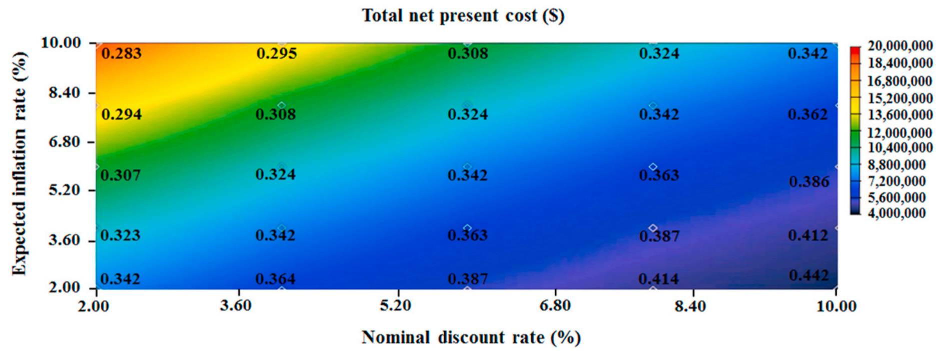

Fig. 21. Effect of the expected inflation rate and discount rate on NPC and LCOE of HRECS.

improving the efficiency of PV cells. Technological advancements have led to an increase in the conversion efficiency of the average solar panel from 15% to well over 20%. As of 2022, the most efficient commercial panel is about 22% (Svarc, 2022). The effect of the Biogas generator's minimum load ratio and cell efficiency of solar PV on electricity production per year and LCOE was investigated. As seen in Fig. 22, as cell efficiency increases, the annual electricity production increases with a decreasing trend in the cost of 1 kWh of production. Also, as the minimum load ratio of the Biogas generator increases (except at 50%), the electricity production per year generally increases with a decreasing trend in the cost of 1 kWh of production. At a 30% minimum load ratio, when the PV cell efficiency increases from 15 to 24%, the LCOE of the system slightly decreases by 0.75%, whilst electricity production slightly increases by 0.50%. Also, at a 20% minimum load ratio, when the PV cell efficiency increases from 15% to 24%, the LCOE of the system decreases by 2.66%, whilst electricity production slightly increases by 0.60%. At 17.3% solar PV efficiency, when the minimum load ratio increases from 10 to 40%, the LCOE decreases by 37%, whilst electricity production increases slightly by 0.30%. Similarly, at 24% solar PV efficiency, when the minimum load ratio increases from 10% to 40%, the LCOE decreases by 36%, whilst electricity production slightly increases by 0.32%.

Fig. 23 shows the effect of PV cost ratio and biomass price on the NPC and LCOE of HRECS. It is observed that as the PV cost ratio and biomass price reduces, the system's NPC and LCOE decrease accordingly. At $40/tonne of biomass, when the PV capital cost ratio reduces from 1–0.300, the NPC and LCOE decrease from $4.78-$4.29 million and $0.414-$0.371/kWh, respectively. Similarly, at a PV capital cost ratio of 1, when the biomass price reduces from $80-$0/tonne, the NPC and LCOE of the system decrease from $5.79-$3.77 million and $0.501-$0.326/kWh, respectively. At a PV capital cost ratio of 0.30 and biomass price of $0/tonne, the lowest LCOE and NPC of the analysis are achieved. As technology continues to develop and resources become readily and cheaply available, the capital cost and biomass price are expected to decrease – resulting in a more economically attractive HRECS in the selected location.

The lifetime of the electrolyzer plays a significant role in the project's lifecycle cost; improving the lifetime of the electrolyzers is of great interest within the scientific community to reduce the cost of producing green hydrogen. PEM electrolyzers have a lifetime between 50,000–80,000 h (~6–9 years) currently, and continuous development by 2050 could see an increase to 100,000–120,000 h (IRENA, 2020). Results in the economic section also show that the hydrogen tank contributes to 11% of HRECS costs, only behind solar PV and Biogas generator. Reducing the capital cost of these tanks will provide an opportunity to reduce the current cost of producing 1 kg of hydrogen at HRECS. Fig. 24 shows the effect of electrolyzer lifetime and hydrogen tank capital cost ratio on the LCOH of the project. It is seen that the former is inversely proportional to LCOH, while the latter and LCOH exhibit direct proportionality. At an electrolyzer lifetime of 15 years, when the capital cost ratio of the hydrogen tanks reduces from 1 to 0.3, the LCOH reduces by 7.96%. Similarly, at a hydrogen capital cost ratio of 1, when the lifetime of the electrolyzer increases from 5 to 25 years, the LCOH reduces by 5.62%. Overall, at a hydrogen tank capital cost ratio of 0.03 and electrolyzer lifetime of 25 years, the lowest possible LCOH in the current analysis is achieved, which is 8.92% lower than the initial condition (hydrogen tank capital cost ratio:1 and electrolyzer lifetime: 15 years).

The previous section shows that a grid-connected RE system has better economic feasibility for HRECS compared to a standalone system at the selected location. Hence, a sensitivity analysis for the effect of a grid sale capacity and electricity sell back rate on NPC and LCOE is investigated, as shown in Fig. 25. It is seen that at a sell back rate of $0.08/kWh, when the grid capacity increases from 90 kW to 290 kW, the cost of producing 1 kWh of electricity decreases from $0.218 to $0.204 whiles, whereas NPC decreases from $2.71–2.66 million. Also, at a fixed grid sale capacity (290 kW), when the electricity sell back rate increases from $0.05-$0.09/kWh, the cost of producing 1 kWh of electricity at

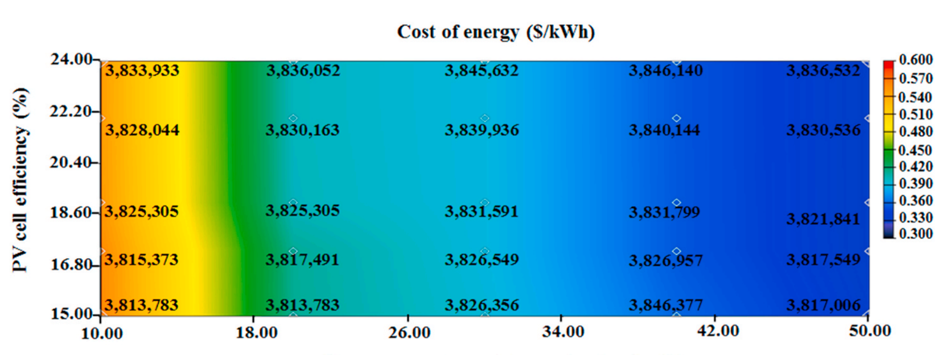

Fig. 22. Effect of generator minimum load ratio and PV cell efficiency on annual electricity production LCOE of HRECS.

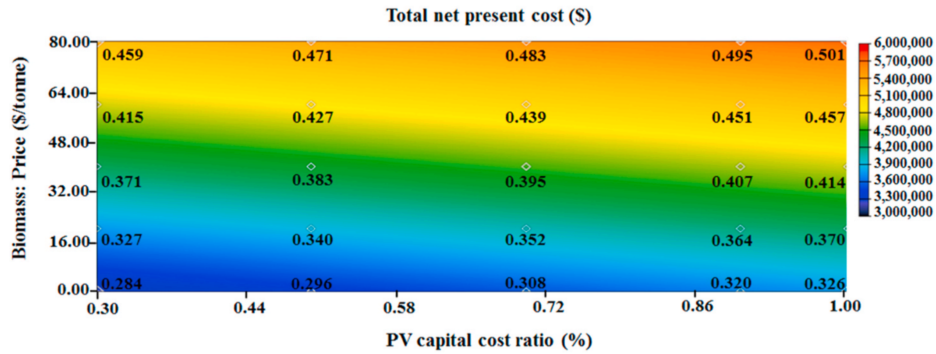

Fig. 23. Effect of PV capital cost ratio and biomass price on NPC and LCOE of HRECS.

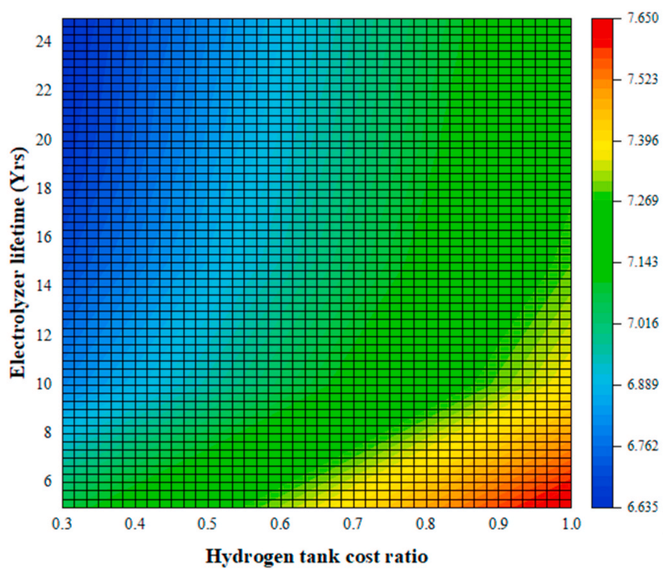

Fig. 24. Effect of hydrogen tank cost ratio and electrolyzer lifetime on LCOH of HRECS.

HRECS decreases from $0.207 to $0.203 with a decrease in NPC from $2.71 million to $2.65 million. For the various pairings (grid sale capacity and electricity sell back rate) considered in this analysis, the pairing with the least LCOE is a sell back rate of $0.09/kWh and grid sale capacity of 290 kW. This pair also produces the least NPC amongst all options considered.

In summary, the results reported in this section show that the initial results are sound and consistent with real-life scenarios. As development and research progress, the techno-economic feasibility of HRECS will become more attractive than in the present state. Optimal components' cost, fuel cost, and components' efficiency and lifetime are viable solutions for improving the feasibility of a standalone hybrid RE system for HRECS in Sub-Saharan Africa.

8. Conclusion

As electric vehicles adoption continues to surge globally, some of the main setbacks to this growth relates to the availability of charging stations, associated energy demand and pressure on grid power, and drawing 100% of the required power from a fossil fuel-dominated grid. A well-established solution to these barriers is the provision of charging stations powered by standalone or grid-connected renewable energy systems. Using Sub-Saharan Africa as a case study, the current work, presents a new station for simultaneously charging and refuelling 100 battery and fuel cell EVs, respectively. The main findings in the present study are summarized below.

Out of six possible hybrid systems considered, the BG + PV system presented the most techno-economic and environmental feasibility. This winning system produces 3.9 GWh/yr and 55.6 tonnes/yr of electricity and hydrogen, respectively. The NPC, LCOE, LCOH, and payback period was recorded as $6.53 million, $0.52/kWh, $9.09/kg and ~8 years, respectively. Approximately, 4 tonnes of GHG emissions are produced by the HRECS each year, and the amount of CO_2 emissions avoided from replacing 100 gasoline vehicles with the proposed EVs has been obtained as 460 tons/yr. By using the BG + PV system for electricity generation, 1362 tonnes of CO_2 emissions are avoided annually from HRECS as opposed to drawing the required power from a 100% fossil fuel system. A comparative analysis of grid-tied versus standalone system revealed that the former presented relatively superior economic characteristics to the latter but environmental feasibility is better in a standalone system.

Finally, based on the results of sensitivity analysis, some key policy initiatives could be implemented to increase the feasibility and adoption rate of EVs in the region. Policies should mainly target reducing the components' cost through subsidizing importation of components,

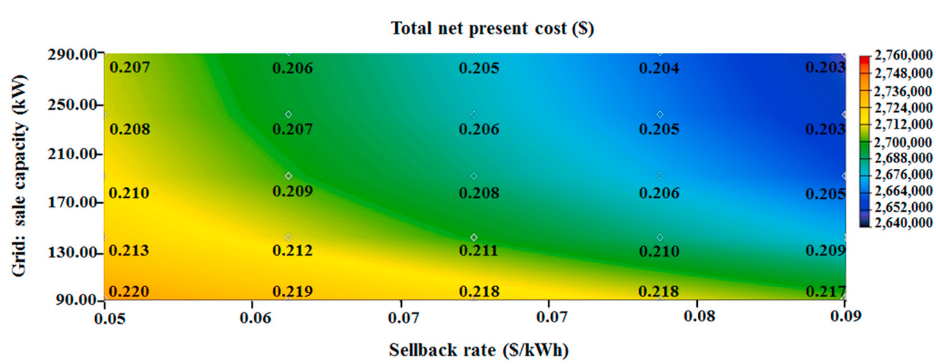

Fig. 25. Effect of grid sale capacity and electricity sell back rate on NPC and LCOE of HRECS.

cutting down associated taxes, and encouraging local manufacturing of components. As incentives are put in place to encourage EVs, there should be a simultaneous pricing mechanism such as carbon pricing in place for conventional vehicles and their fuels which are rather costly. The current work has been successfully modelled for SSA and key contributions have been made to the sustainable development goals, including goals 7, 8, 11, and 13. Future studies in the region will consider other transport sectors, especially the shipping industry. Also, other RE systems such as hydrokinetics could be added to the investigation. Co-generation of different biomass resources such as MSW, animal waste, and agricultural waste could be added to the setup to increase the efficiency and power generation of the Biogas generator. Alternatively, due to the carbon content of biomass, a hydrogen fuel cell could be integrated with the PV system to supplement the electricity production and serve the system at night, on cloudy or rainy days alongside the system batteries, at zero GHG emissions. Moreover, when available, future works could consider real demand profiles for EV charging and refuelling in the selected locality which is the research limitation regarding the present work.

CRediT authorship contribution statement

Jeffrey Dankwa Ampah: Writing – original draft, and Preparation, Software, Investigation, Formal analysis, Writing – review & editing. **Sandylove Afrane:** Data curation, Resources, Software. **Ephraim Bonah Agyekum:** Conceptualization, Methodology. **Humphrey Adun:** Writing – review & editing, Validation. **Abdulfatah Abdu Yusuf:** Validation, Writing – review & editing. **Olusola Bamisile:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jclepro.2022.134238.

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